Hyperspectral Anomaly Detection and Unmixing via Joint Optimization and Deep Interactive Network

Xu He, Yingqian Wang, *Member, IEEE*, Qiang Ling, Kun Li, Yihang Luo, and Shilin Zhou

***Abstract*—Hyperspectral unmixing (HU) and hyperspectral anomaly detection (HAD) are two momentous tasks in remote sensing and Earth observation, where HU identifies material composition and abundance and HAD locates rare targets against complex backgrounds. Existing approaches predominantly follow a sequential unmixing-then-detection paradigm, in which detection performance is inherently constrained by the fidelity and stability of the preceding unmixing stage, leading to suboptimal detection synergy. To address this limitation, we propose a joint optimization and deep interactive network (JODINet) that unifies HU and HAD within a single end-to-end framework. Departing from cascade architectures, the JODINet employs a weight-sharing encoder followed by a parallel background decoder for anomaly discrimination and a spectral decoder for abundance estimation. Central to our design is an abundance-background interaction module (ABIM) that explicitly models the latent dependency between abundance maps and background reconstruction, enabling bidirectional refinement across both branches. Furthermore, a learnable endmember learning module (ELM) is incorporated to precisely estimate background spectral signatures, mitigating the instability common in unsupervised unmixing. By jointly optimizing both tasks, the proposed framework establishes a mutual feedback mechanism that improves background estimation and enhances detection performance, while anomaly-driven cues guide more accurate spectral decomposition. Extensive experiments on real-world HAD and HU datasets demonstrate that the proposed JODINet consistently outperforms state-of-the-art approaches in both anomaly detection accuracy and unmixing precision.**

***Index Terms*—Endmember learning module (ELM), hyperspectral anomaly detection (HAD), hyperspectral unmixing (HU), joint optimization and deep interactive.**

I. Introduction

HYPERSPECTRAL image (HSI) is actually characterized by its 3-D data structure, typically encompassing dozens to hundreds of contiguous spectral bands spanning the visible to infrared spectrum. This fine spectral resolution provides far richer diagnostic information than conventional RGB or multispectral imagery [1], [2], [3]. Consequently, HSI has become an indispensable data source in a wide range of remote sensing and Earth observation applications, including fine-grained land-cover classification [4], [5], [6], [7], [8], [9], [10] and subtle target detection [11], [12], [13], [14]. However, due to the inherent tradeoff between spectral and spatial resolution imposed by sensor optics, mixed pixels, where a single pixel integrates the spectral response of multiple distinct materials, are pervasively present in HSIs. This spectral mixing phenomenon severely degrades the performance of high-level semantic interpretation tasks that rely on pure material signatures.

Hyperspectral unmixing (HU) has emerged as a fundamental technique to address the mixed pixel problem by decomposing each observed spectrum into a set of constituent pure spectral signatures, termed endmembers, and their corresponding fractional proportions, termed abundances [15]. Depending on the availability of a priori spectral libraries, HU methodologies can be broadly categorized into supervised and unsupervised approaches. In practice, acquiring a comprehensive and scene-specific spectral library is exceptionally challenging due to atmospheric interference, illumination variations, and temporal changes. As a result, unsupervised unmixing, which estimates endmembers and abundances directly from the image data without external libraries, has garnered significant research attention. Yet, most unsupervised HU algorithms rely on minimizing the reconstruction error under certain constraints, which does not inherently guarantee that the extracted abundances align with the actual physical distribution of materials. Without ground-truth guidance, the resulting unmixing outputs are often unstable and imprecise, making their quality difficult to evaluate in isolation.

In the HSI processing chain, HU is conventionally treated as a preprocessing step, with its output abundance maps intended to facilitate downstream high-level semantic tasks. A typical and highly relevant example is hyperspectral anomaly detection (HAD), which aims to identify rare pixels whose spectral characteristics deviate significantly from the prevailing background distribution. Intuitively, unmixing can provide a more discriminative feature space for anomaly detection: background materials correspond to dominant endmembers with high abundances, whereas anomalies may manifest as low-abundance or off-model spectral components. This paradigm of leveraging HU for HAD has been explored in several prior works [16], [17], [18], [19], [20]. Nevertheless, a critical limitation persists: the improvement in HAD performance is

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Xu He is with Information Support Force Engineering University, Chongqing 400035, China, and also with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: hexu_710@nudt.edu.cn).

Yingqian Wang, Qiang Ling, Yihang Luo, and Shilin Zhou are with the College of Electronic Science and Technology, National University of Defense Technology, Changsha 410073, China (e-mail: wangyingqian16@nudt.edu.cn; lq910131@163.com; luoyihang@nudt.edu.cn; slzhou@nudt.edu.cn).

Kun Li is with the College of Big Data and Information Engineering, Guizhou University, Guiyang 550025, China (e-mail: lik@gzu.edu.cn).

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entirely contingent upon the accuracy and stability of the upstream unmixing stage. Conventional HU algorithms are designed and optimized independently of the HAD objective, guided solely by reconstruction error minimization. The extracted abundance features may not possess the optimal separability required to distinguish anomalies from background clutter. In such a sequential unmixing-then-detection framework, errors and uncertainties introduced during unmixing propagate irreversibly into the detection phase, leading to suboptimal synergy and, in some cases, even degraded detection performance.

To overcome this inherent limitation, it is imperative to move beyond the decoupled, cascading paradigm and establish a more integrated framework in which HU and HAD can mutually guide and constrain each other. Ideally, the high-level task (detection) should serve as a meaningful evaluative criterion for the low-level task (unmixing), steering the decomposition toward features that enhance target–background discrimination rather than merely minimizing spectral reconstruction error. However, the investigation of such joint optimization frameworks, particularly within the context of deep learning, remains largely underexplored. At present, a few studies have attempted to connect unmixing with semantic tasks. For instance, Qu et al. [16] and Song et al. [17] performed HAD in the abundance domain using precomputed unmixing results, which are fundamentally sequential. The unmixing and detection processes are conducted separately according to their own objective functions, failing to establish a bidirectional feedback mechanism that could mitigate cumulative errors. Recently, Liu et al. [21] introduced an end-to-end deep learning architecture named UADNet for HAD and HU by jointly optimizing feature extraction and clustering. However, UADNet is actually identified as a sequential unmixing-then-detection model rather than a true joint unmixing-detection system, where the performance of subsequent detection is influenced by the effect of front-end unmixing.

In this article, we investigate a novel perspective for synergistically combining HU and HAD within a unified, end-to-end deep learning framework. Fig. 1 illustrates the bidirectional information flow between HU and HAD in the proposed parallel framework, contrasted with conventional sequential pipelines. In the unmixing stream, the estimated abundance maps in Fig. 1(b) inherently encode material-level spatial distributions, which serve as saliency priors for anomaly localization: regions exhibiting abrupt compositional changes or spectral dissimilarity to neighboring background abundances provide critical cues for identifying potential anomalous targets. Conversely, in the detection stream, the residual map [Fig. 1(d)] derived from background reconstruction highlights pixels that deviate from the learned background model. These residuals not only flag anomalies but also indicate where the abundance estimation, which fundamentally models background constituent proportions, may be unreliable due to the presence of outliers. Specifically, the residual magnitude inversely correlates with the confidence of background abundance estimation, thereby offering a feedback signal to refine the unmixing process in anomaly-affected regions. This mutual guidance mechanism enables the joint framework to

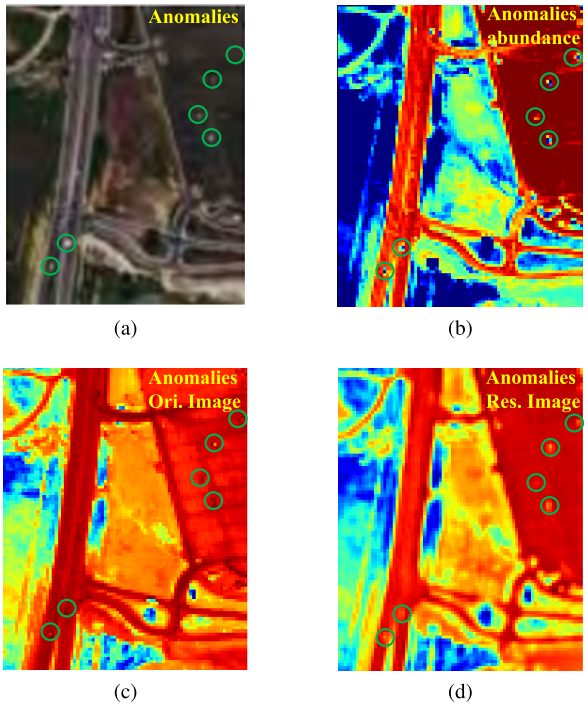


Fig. 1. Schematic of the interactive mechanisms between HAD and HU. (a) Pseudocolor image of HYDICE dataset. (b) HSI unmixing abundance maps. (c) HSI original heatmap. (d) HSI reconstruction residual maps.

simultaneously improve the fidelity of abundance maps and the discriminability of the detection map, circumventing the error accumulation inherent in sequential unmixing-then-detection pipelines, e.g., UADNet, where unmixing inaccuracies irreversibly contaminate subsequent background modeling. This raises a pivotal question: 1) how can we design such a parallel, mutually interactive architecture to fully exploit this synergy and eliminate the error accumulation endemic to sequential processing? and 2) is it feasible to endow the latent feature space of a detector with physically grounded abundance semantics, thereby enabling a joint optimization framework that simultaneously advances both detection accuracy and unmixing precision?

To address the above issues, we propose a joint optimization and deep interactive network (JODINet) that unifies HU and anomaly detection within a single, coherent end-to-end architecture. Departing fundamentally from traditional cascade structures in UADNet, JODINet employs a weight-sharing encoder network to project the input HSI into a compact latent representation. From this shared representation, two parallel decoding branches are instantiated: a background decoder dedicated to reconstructing the background scene for anomaly discrimination and a spectral decoder tasked with estimating endmembers and generating physically meaningful abundance maps. A pivotal innovation of our framework is the abundance-background interaction module (ABIM), which explicitly models the latent dependencies between the estimated abundance distributions and the reconstructed background. This module facilitates adaptive, bidirectional refinement; accurate background reconstruction aids in purifying the extracted endmembers, while well-constrained abundance maps guide

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the background decoder to better suppress clutter and highlight anomalous deviations. Furthermore, we incorporate a learnable endmember learning module (ELM) to precisely estimate the spectral signatures of background materials directly from the data, thereby mitigating the instability commonly associated with purely unsupervised endmember extraction algorithms. By jointly optimizing the reconstruction losses of both decoders alongside interaction constraints, JODINet establishes a mutual feedback mechanism wherein improved background estimation enhances anomaly detection performance, while detection-driven cues conversely guide the unmixing branch toward more discriminative spectral decomposition.

In summary, the major contributions of this work are fourfold.

1) We propose JODINet, a novel deep network architecture that unifies unsupervised HU and anomaly detection in a fully end-to-end manner. Unlike prior sequential approaches, our framework enables joint optimization of both tasks through a shared encoder and a carefully designed multiobjective loss function, allowing them to mutually enhance one another and effectively eliminating the cumulative error of cascade pipelines.
2) To fully exploit the intrinsic relationship between spectral decomposition and background estimation, we design a dedicated interaction module. This module mines the structural correspondence between abundance maps and background reconstructions, enabling adaptive cross-task feature refinement. This bidirectional information flow ensures that the unmixing branch produces features that are not only physically accurate but also semantically discriminative for anomaly detection.
3) We introduce a learnable module within the spectral decoder that dynamically estimates background endmembers during training. This design circumvents the sensitivity of traditional unsupervised unmixing to initialization and noise, leading to more stable and precise endmember and abundance estimation that directly benefit the downstream detection process.
4) We conduct extensive experiments on multiple HAD and HU datasets to validate the superiority of JODINet as compared to existing methods.

The remainder of this article is organized as follows. Section II reviews related work on HU and anomaly detection, establishing the technical background. Section III provides a detailed exposition of the proposed JODINet architecture, including the encoder–decoder structure, the ABIM, and the joint optimization strategy. Section IV presents extensive experimental evaluations on both real-world and benchmark hyperspectral datasets, along with comparative analyses against state-of-the-art methods. Finally, Section V concludes this article with a discussion of our findings and potential directions for future research.

II. Related Work

In this section, we first briefly review the related work of our study, including HU and HAD.

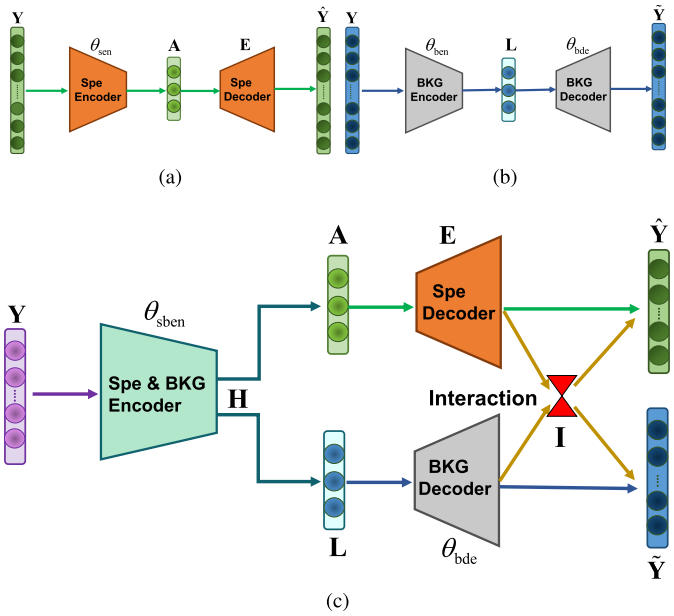


Fig. 2. (a) Schematic of the AE-based unmixing network for HU task. (b) Schematic of the background reconstruction-based anomaly detection network. (c) Schematic of the anomaly detection and unmixing via JODINet.

A. Hyperspectral Unmixing

HU decomposes a mixed pixel spectrum into a set of pure material signatures (endmembers) and their corresponding fractional abundances. The linear mixture model (LMM) is the most widely adopted formulation due to its physical interpretability [22]. Under the LMM, an observed hyperspectral image $\mathbf{Y} \in \mathbb{R}^{B\times P}$ (with B spectral bands and P pixels) is modeled as

$$\mathbf{Y} = \mathbf{E}\mathbf{A} + \mathbf{N} \tag{1}$$

where $\mathbf{E} \in \mathbb{R}^{B\times R}$ (with R endmembers) denotes the endmember matrix, $\mathbf{A} \in \mathbb{R}^{R\times P}$ is the abundance matrix, and $\mathbf{N}$ represents additive noise or other unmodeled factors. The model is subject to the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC) [23]. Based on this principle, traditional unmixing approaches include geometry-based techniques [24], [25], statistical inference methods [26], [27], sparse regression frameworks [28], [29], and nonnegative matrix factorization (NMF) [30]. However, these methods rely heavily on handcrafted features and explicit model assumptions, which often limit their adaptability to complex real-world scenes.

Autoencoders (AEs) have emerged as a powerful paradigm for blind HU [31]. In the AE-based formulation, the encoder $Enc(\cdot)$ compresses the input spectral vector into a low-dimensional latent representation that is physically interpreted as the abundance vector, while the decoder $Dec(\cdot)$ reconstructs the input using a linear transformation whose weights correspond to the endmember matrix. Following the AE unmixing paradigm, numerous unmixing algorithms, depicted in Fig. 2(a), adopt a specialized encoder–decoder architecture to extract the abundance matrix $\mathbf{A}$ and endmember matrix $\mathbf{E}$. The input HSI $\mathbf{Y}$ is first processed by a spectral encoder parameterized by θ_{sen}. This encoder maps the spectral data into

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a low-dimensional abundance space that intrinsically satisfies the physical constraints of ANC and ASC. The encoding operation is expressed as

$$\mathbf{A} = Enc_{\text{spe}}(\mathbf{Y}; \theta_{\text{sen}}) = \sigma(\mathbf{W}_{\text{sen}}\ \mathbf{Y} + \mathbf{b}_{\text{sen}}) \tag{2}$$

where $\sigma(\cdot)$ denotes a nonlinear activation function (e.g., ReLU followed by a channel-wise softmax to enforce the sum-to-one property) and $\mathbf{W}_{\text{sen}}$ and $\mathbf{b}_{\text{sen}}$ are the trainable weights and biases. Subsequently, a spectral decoder with parameters θ_{sde} reconstructs the HSI from the abundance matrix $\mathbf{A} \in \mathbb{R}^{R\times P}$. The decoder is constrained to be a linear transformation without a bias term, and its weight matrix directly corresponds to the estimated endmember matrix $\mathbf{E} \in \mathbb{R}^{B\times R}$. The reconstructed spectrum $\hat{\mathbf{Y}}$ is given by

$$\hat{\mathbf{Y}} = Dec_{\text{spe}}(\mathbf{A}; \theta_{\text{sde}}) = \mathbf{EA} \tag{3}$$

where the nonnegativity constraint $\mathbf{E}_{br} \geq 0$ is enforced on the decoder weights. The module is trained by minimizing the spectral reconstruction loss $\|\mathbf{Y} - \mathbf{EA}\|_F^2$ with additional regularization to guarantee $\mathbf{A}_{rp} \geq 0$ and $\sum_{r=1}^{R} \mathbf{A}_{rp} = 1$ for each pixel p. Palsson et al. [32] investigated the impact of various objective functions on deep AE unmixing performance. To incorporate spatial context, a convolutional AE (CNNAEU) was proposed in [33] to boost the unmixing performance. Jin et al. [34] introduced an adversarial AE (AAE) to better align the abundance distribution with a prior. Su et al. designed a deep AE network (DAEN) [35] that employs stacked AEs for robust initialization and a variational AE to simultaneously estimate endmembers and abundances while handling outliers and low signal-to-noise ratios. Furthermore, Wang et al. [36] proposed a deep AE-based nonlinear unmixing framework in which a decoder comprising a linear endmember extraction component and a nonlinear mixing module learns complex spectral interactions without relying on an explicit physical model. Rasti et al. [37] developed a minimum simplex convolutional network (MiSiCNet) that incorporates spatial information via convolutional filters and geometric constraints through a simplex volume penalty, enabling robust blind unmixing even in the absence of pure pixels. Shi et al. [38] presented a probabilistic generative model based on a variational AE that jointly estimates abundances and pixel-wise endmembers while capturing arbitrary endmember distributions to address spectral variability. Gao et al. [39] introduced a cycle-consistency unmixing network (CyCU-Net) that cascades two AEs with a self-perception loss to preserve high-level semantic information and improve unmixing accuracy. More recently, Gao et al. [40] proposed a spatial–spectral adaptive fusion network (SSAF-Net) that utilizes cascaded encoders with channel and spatial attention mechanisms, together with a deep generative model, to account for endmember variability and achieve effective local–global information fusion. Yu et al. [41] developed a multistage convolutional AE network (MSNet) that fuses multiscale features to preserve local details and global context. In addition, Li et al. [42] formulated a multitemporal latent dynamical unmixing framework based on neural ordinary differential equations, where abundance evolution is modeled via dynamical discretization with theoretical guarantees of consistency, convergence, and stability. Furthermore, Li et al. [43] presented a copula-guided temporal dependency method for multitemporal HU that explicitly models joint distributions across time phases to capture material dynamics and guide the unmixing process. Meanwhile, Gao et al. [44] designed a reversible generative network that learns an explicit, invertible distribution for endmembers using flow-based transformations, thereby addressing spectral variability in HU while avoiding mode collapse and blurring.

B. Hyperspectral Anomaly Detection

HAD aims to identify pixels whose spectral characteristics deviate significantly from the predominant background distribution without relying on a priori target signatures. Existing HAD methods can be broadly categorized into traditional approaches and deep learning-based approaches. Traditional HAD methods primarily rely on statistical assumptions or data representation models. The seminal Reed–Xiaoli (RX) detector [11] assumes a multivariate Gaussian background and employs the Mahalanobis distance to flag anomalies. Its variants include SSRX [45], LFRX [46], and FrFE-RX [47]. Representation-based models such as collaborative representation-based detector (CRD) [48], [49], sparse representation-based detector (SRD) [50], [51], and low-rank representation (LRR) [52], [53], [54] will separate anomalies from the background by exploiting linear representation or low-rank properties. While effective in controlled settings, these methods are often constrained by strict distributional priors or computationally expensive iterative optimization.

Deep learning-based detectors circumvent these limitations by learning to reconstruct the background in a self-supervised manner. AEs are widely employed under the premise that anomalies exhibit larger reconstruction errors. As illustrated in Fig. 2(b), the reconstruction-based anomaly detection adheres to the deep reconstruction-based paradigm. The input HSI $\mathbf{Y}$ is passed through a background encoder with parameters θ_{ben}, which compresses the spectral–spatial information into a compact latent background representation $\mathbf{L} \in \mathbb{R}^{D\times P} (D \ll B)$

$$\mathbf{L} = Enc_{\text{bkg}}(\mathbf{Y}; \theta_{\text{ben}}) = f_{\text{ben}}(\mathbf{Y}). \tag{4}$$

A background decoder parameterized by θ_{bde} then maps the latent background code $\mathbf{L}$ back to the original spectral domain, yielding a background-only reconstruction $\tilde{\mathbf{Y}}$

$$\tilde{\mathbf{Y}} = Dec_{\text{bkg}}(\mathbf{L}; \theta_{\text{bde}}). \tag{5}$$

The module is optimized by minimizing the background reconstruction loss $\mathcal{L}_{\text{bkg}} = \|\mathbf{Y} - \tilde{\mathbf{Y}}\|_F^2$. During inference, the anomaly score for the p-th pixel is computed as the Euclidean norm of the reconstruction residual

$$s_p = \left\|\mathbf{y}_p - \tilde{\mathbf{y}}_p\right\|_2. \tag{6}$$

Since the network is trained predominantly on background pixels, it learns to reconstruct them with high fidelity, whereas anomalies induce substantially larger residuals. Notable works include Auto-AD [55] with a U-shaped fully convolutional AE, RGAE [56] with graph AE technology, DFAE [57] that

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processes high- and low-frequency components in parallel, and WeaklyAD [58], which leverages adversarial learning to amplify anomaly residuals. In addition, Jiang et al. [59] devised an unsupervised low-rank embedded network (LREN) in which the latent representation layer of AE is integrated with a Gaussian mixture model to enhance feature discrimination. In particular, a blind-spot self-supervised learning network was proposed by Gao et al. [60] and Wang et al. [61], [62], wherein a dedicated anomaly-overshadowing architecture enables the generation of highly accurate reconstruction error maps. More recently, Vision Transformers (ViTs) [63], [64], [65] and Mamba-based architectures [66], [67], [68] have emerged as powerful alternatives for modeling long-range spatial–spectral dependencies with enhanced computational efficiency. In addition, domain adaptation has been increasingly applied in remote sensing object detection to mitigate performance degradation caused by domain shifts across different sensors, scenes, or imaging modalities. For example, Zhang et al. [69] developed a foreground instance enhancement network that integrates multigranularity feature alignment with label filtering to suppress background interference and emphasize foreground features for unsupervised domain adaptation object detection in remote sensing imagery. Wang et al. [70] introduced an anchor-free domain adaptation framework for SAR ship detection, which aligns keypoint heatmaps and bounding box regressions across optical and SAR domains to enable robust cross-modality knowledge transfer. In the HAD field, Hou et al. [71] designed a cross-scene adversarial learning framework that leverages Gaussian mixture modeling and domain adaptation to transfer background knowledge from a source scene to a target scene, thereby improving HAD without requiring prior anomaly information.

III. Proposed Methodology

In this section, we first introduce the basic mechanism of JODINet for anomaly detection and unmixing tasks. Subsequently, we introduce a particular explanation of the proposed JODINet, covering the overall architecture and specific module design.

A. Basic Mechanism

The principal contribution of this work is the synergistic integration of the unmixing and anomaly detection streams, as illustrated in Fig. 2(c). This joint framework is designed to exploit the intrinsic relationship between spectral mixture physics and background modeling, thereby enhancing the performance of both tasks simultaneously.

1) Shared Encoder: The input HSI $\mathbf{Y}$ is first processed by a unified Spe&BKG Encoder parameterized by θ_{sben} , which extracts a common feature representation $\mathbf{H}$

$$\mathbf{H} = Enc_{\text{shared}}(\mathbf{Y}; \theta_{\text{sben}}). \tag{7}$$

The shared representation $\mathbf{H}$ is subsequently routed into two parallel branches with an explicit interaction module.

2) Spectral Unmixing: The features $\mathbf{H}$ are projected by the spectral branch to produce the abundance matrix $\mathbf{A}$. The spectral decoder reconstructs the spectral data via a linear transformation whose weights form the endmember matrix $\mathbf{E}$

$$\hat{\mathbf{Y}} = \mathbf{EA}. \tag{8}$$

This branch is supervised by the spectral reconstruction loss $\mathcal{L}_{\text{HU}} = \|\mathbf{Y} - \mathbf{EA}\|_F^2$ with physical constraints (ANC, ASC, and $\mathbf{E} \geq 0$).

3) Background Reconstruction: In parallel, the shared features $\mathbf{H}$ are represented by background-specific cues. Specifically, the latent features $\mathbf{L}$ serve as the input to background decoder

$$\tilde{\mathbf{Y}} = Dec_{\text{bkg}}(\mathbf{L}; \theta_{\text{bde}}). \tag{9}$$

This branch is supervised by the background reconstruction loss $\mathcal{L}_{\text{HAD}} = \|\mathbf{Y} - \tilde{\mathbf{Y}}\|_F^2$.

4) Synergistic Training Objective: The overall network is trained end-to-end by minimizing a composite loss function

$$\mathcal{L}_{\text{total}} = \underbrace{\|\mathbf{Y} - \mathbf{EA}\|_F^2}_{\text{HU Reconstruction}} + \lambda \underbrace{\|\mathbf{Y} - \tilde{\mathbf{Y}}\|_F^2}_{\text{HAD Reconstruction}} + \mathcal{R}(\mathbf{A}, \mathbf{E}) \tag{10}$$

where λ balances the two objectives and $\mathcal{R}(\cdot)$ denotes the physical regularizers. The shared encoder, coupled with the interaction module, ensures that the unmixing branch imposes a physically structured regularization on the latent space, preventing the background encoder from learning to reconstruct anomalies. Conversely, the background suppression objective encourages the encoder to filter out sparse outliers, leading to more robust endmember estimation. This closed-loop synergy yields a unified model capable of simultaneous endmember extraction, abundance estimation, and precise anomaly localization in complex hyperspectral scenes. In addition, as presented in Fig. 2(c), the interaction mechanism enables the bidirectional flow of information, wherein the refined representation $\mathbf{I}$ simultaneously guides the Spe Decoder toward physically consistent abundance estimation and the BKG Decoder toward effective background suppression.

B. Overall Network Architecture

As illustrated in Fig. 3, the proposed framework adopts a dual-branch encoder–decoder architecture that simultaneously performs HU and anomaly detection. The input HSI is first partitioned into overlapping spatial patches of size $P \times P \times B$ via a sliding window operation, ensuring that local spatial–spectral context is preserved for subsequent feature extraction. Each patch $h_i \in \mathbb{R}^{P \times P \times B}$ is then processed by a shared Spe&BKG Encoder, which extracts hierarchical features through consecutive 3×3 convolutional layers with LeakyReLU, batch normalization, and dropout, ultimately yielding a compact latent representation of dimension $P \times P \times 64$. From this shared representation, two specialized decoders operate in parallel. The upper branch denotes the spectral decoder that reconstructs the input via a linear transformation $\hat{h}_i = \mathbf{EA}$, where the decoder weights correspond to the endmember matrix $\mathbf{E}$ and the output constitutes the abundance map $\mathbf{A}$. The abundance map $\mathbf{A}$ is obtained from the encoder output

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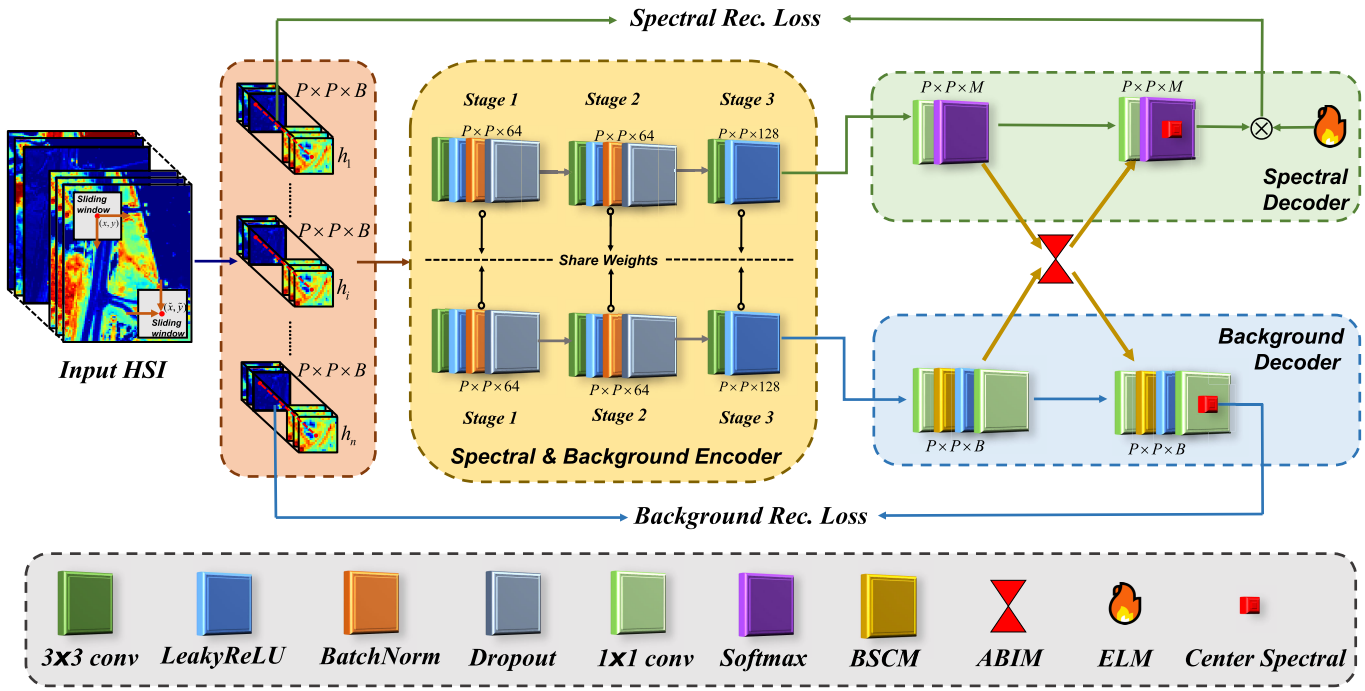


Fig. 3. Framework of JODINet with shared spectral and background encoder and the corresponding spectral decoder and background decoder. The proposed JODINet comprises the BSCM, ABIM, and the ELM.

through a softmax activation, ensuring compliance with the abundance nonnegativity and sum-to-one constraints. Critically, the endmember matrix $\mathbf{E}$ is not fixed but is dynamically learned through a dedicated ELM integrated within the spectral decoder branch. The ELM adaptively refines the endmember signatures during training, allowing the model to capture spectral variations and material-specific characteristics inherent in the scene, thereby improving both the fidelity of spectral reconstruction and the physical interpretability of the unmixing process. The lower branch represents the Background Decoder that reconstructs background $\tilde{h}_i$, from which anomalies are identified via reconstruction residuals. These two streams are self-supervised by spectral reconstruction and background reconstruction losses, ensuring that the joint optimization yields physically meaningful abundance estimation and anomaly detection maps. In addition, to suppress anomalies during background learning, the background spectral contrastive module (BSCM) is integrated into the background branch. The BSCM explicitly contrasts the spectral signatures of potential anomalies against their surrounding background context. The interaction between unmixing and detection is realized through the ABIM.

C. Background Spectral Contrastive Module

The BSCM is designed to explicitly suppress anomaly signatures by exploiting the inherent spectral discrepancy between a pixel's central spectral and its surrounding background spectral. As depicted in Fig. 4, it computes query–key attention between convolutional projections of the feature map, followed by a softmax operation and value projection, thereby enhancing background homogeneity while attenuating outlier spectral that deviate from local neighborhood consistency.

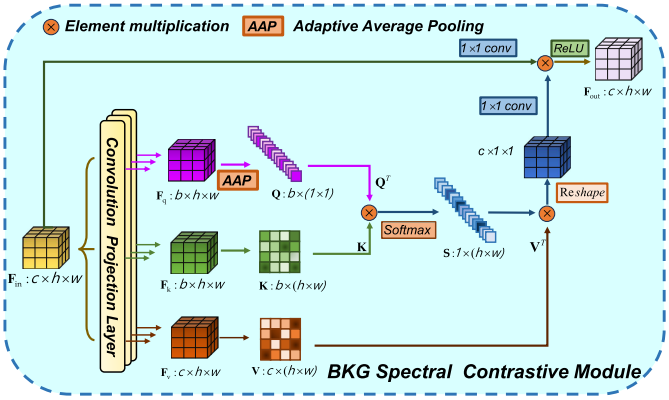


Fig. 4. Network design of the proposed BSCM.

Specifically, given an input feature patch $\mathbf{F}_{\text{in}} \in \mathbb{R}^{c\times h\times w}$, two parallel 1×1 convolutions first project it to a reduced bottleneck dimension $b = c/2$, yielding query/key tensor $\mathbf{F}_{\text{q}}, \mathbf{F}_{\text{k}} \in \mathbb{R}^{b\times h\times w}$ and a value tensor $\mathbf{F}_{\text{v}} \in \mathbb{R}^{c\times h\times w}$. To capture the representative central spectral of each patch, $\mathbf{F}_{\text{q}}$ undergoes adaptive average pooling (AAP), and the pooled result is flattened to form the query matrix $\mathbf{Q} \in \mathbb{R}^{(1\cdot 1)\times b}$. Meanwhile, the original unpooled $\mathbf{F}_{\text{k}}$ is flattened to serve as the key matrix $\mathbf{K} \in \mathbb{R}^{b\times(h\cdot w)}$, which encodes the background spectral context of the local neighborhood. A scaled dot-product attention map $\mathbf{S} = \text{softmax}(\mathbf{Q}^{\top}\mathbf{K}/\sqrt{b}) \in \mathbb{R}^{1\times(h\cdot w)}$ is then computed to measure the spectral similarity between each central query and its surrounding background keys. Background pixels are spectrally consistent with their background and receive high attention weights, whereas anomalous pixels, whose central spectra deviate from the local background distribution, are

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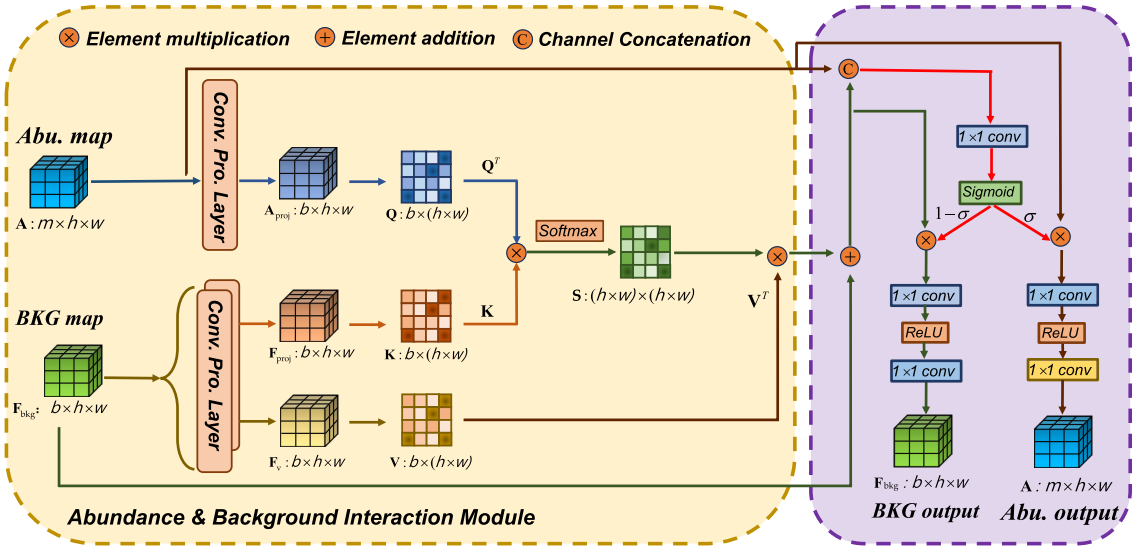


Fig. 5. Schematic of the proposed ABIM, which is used for HAD and HU interaction.

naturally suppressed. The attention map is subsequently applied to the flattened value matrix $\mathbf{V}^T \in \mathbb{R}^{(h\cdot w)\times c}$, and after $1\times$ convolution, ReLU activation, and element-wise multiplication with the original input $\mathbf{F}_{\text{in}}$, the module outputs a refined feature map $\mathbf{F}_{\text{out}} \in \mathbb{R}^{c\times h\times w}$ wherein background homogeneity is reinforced and outlier responses are substantially attenuated, thereby preventing the background decoder from learning to reconstruct anomalies.

D. Abundance-Background Interaction Module

The ABIM establishes a bidirectional cross-attention mechanism between the unmixing and detection streams, enabling the abundance map to guide background feature refinement while simultaneously allowing the background reconstruction context to inform abundance estimation. As illustrated in Fig. 5, the module receives two inputs: the abundance map $\mathbf{A} \in \mathbb{R}^{m\times h\times w}$ from the spectral encoder, where m denotes the number of endmembers, and the background feature map $\mathbf{F}_{\text{bkg}} \in \mathbb{R}^{b\times h\times w}$ from the shared encoder, where b is the feature channel dimension. Both inputs first pass through dedicated convolutional projection layers, which align their channel dimensions to a common bottleneck size b, yielding projected tensors $\mathbf{A}_{\text{proj}} \in \mathbb{R}^{b\times h\times w}$ and $\mathbf{F}_{\text{proj}} \in \mathbb{R}^{b\times h\times w}$. These projected features are then flattened to form matrices of shape $b \times (h \cdot w)$. The interaction core employs a cross-attention block wherein the abundance-derived features serve as the query matrix $\mathbf{Q} \in \mathbb{R}^{b\times(h\cdot w)}$, while the background-derived features serve as both the key matrix $\mathbf{K} \in \mathbb{R}^{b\times(h\cdot w)}$ and the value matrix $\mathbf{V} \in \mathbb{R}^{b\times(h\cdot w)}$. A scaled dot-product attention map $\mathbf{S} = \text{Softmax}(\mathbf{Q}^\top\mathbf{K}/\sqrt{b}) \in \mathbb{R}^{(h\cdot w)\times(h\cdot w)}$ is then computed, capturing the spatial–spectral correspondence between abundance composition and background spectral context. These regions with coherent background abundances will yield high attention scores, whereas anomalies induce mismatched responses. The attended representation $\mathbf{SV}^T$ is then added element-wise to $\mathbf{F}_{\text{bkg}}$ via a residual connection and concatenated with $\mathbf{A}$ along the channel dimension. This fused feature is processed by consecutive 1×1 convolutions and sigmoid activations to produce a gating weight map $\boldsymbol{\sigma} \in [0, 1]^{b\times(h\cdot w)}$. The gate $\boldsymbol{\sigma}$ is subsequently applied in a dual-branch manner, which is multiplied element-wise with the original abundance map $\mathbf{A}$ to enhance anomaly-sensitive abundance responses. The complement gate $(\mathbf{1} - \boldsymbol{\sigma})$ is multiplied with the residual-enhanced background features to suppress anomaly-contaminated regions. Both gated outputs are reshaped back to spatial dimensions, passed through dedicated 1×1 convolutions with ReLU activations, and finally restored to their original channel dimensions. The module thereby produces two refined outputs: a purified abundance map $\tilde{\mathbf{A}} \in \mathbb{R}^{m\times h\times w}$ and an enhanced background feature map $\tilde{\mathbf{F}}_{\text{bkg}} \in \mathbb{R}^{b\times h\times w}$, which are fed into the spectral decoder and background decoder to jointly improve unmixing fidelity and anomaly suppression.

E. Endmember Learning Module

To enhance the accuracy and robustness of our unmixing network, we draw from classical unmixing techniques and initialize the weights $\mathbf{E}$ with the endmembers extracted by VCA [25]. However, the endmembers initialized by VCA often deviate from true material signatures in complex scenes. Therefore, we devised an ELM to adaptively refine $\mathbf{E}$ throughout training to capture scene-specific spectral variations, thereby enhancing endmember estimation accuracy.

As illustrated in Fig. 6, the ELM further refines the initial endmember matrix with the size of $M \times B$ through a series of learnable transformations and a spectral attention mechanism. The module begins with four sequential stages (Stages 1–4), each consisting of a linear layer, a GELU activation function, and spectral normalization (Spe_norm). These stages progressively increase the representational capacity while stabilizing training by controlling the Lipschitz constant of the transformations. After the first two stages, the module applies a spectral attention endmember module, which operates on a feature map of size $M \times C$ (where $C = 256$ represents

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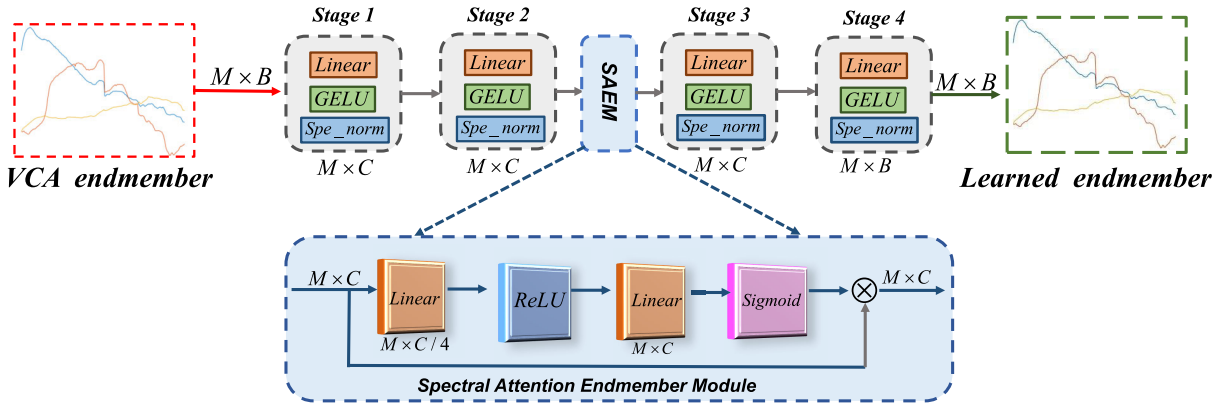


Fig. 6. Overall structure of the proposed ELM, which is used for accurate and dynamic endmember estimation.

the hidden dimension, typically larger than B). The attention block first passes the features through a linear layer that reduces the dimension to $M \times C/4$, followed by a ReLU activation. A second linear layer then projects the features back to $M \times C$, and a sigmoid activation generates attention weights in the range $[0, 1]$. These weights are multiplied element-wise with the original $M \times C$ features (via a residual connection or a gating mechanism) to highlight the most discriminative spectral components while suppressing less informative ones. Specifically, the last stage maps the attended features back to the original spectral dimension, producing a refined endmember matrix of size $M \times B$. In essence, the ELM adds nonlinear depth and a self-attention mechanism that adaptively reweights spectral bands, allowing the network to correct residual errors from the initial VCA estimation, mitigate noise, and emphasize physically meaningful spectral signatures. The entire module remains fully differentiable and can be trained jointly with the rest of the network, making it a powerful extension for high-quality, adaptive endmember extraction in complex hyperspectral scenes.

F. Objective Function

The overall loss function of our proposed network consists of four components: a spectral reconstruction loss for unmixing, a background reconstruction loss, an endmember smoothness regularization loss, and an endmember orthogonality loss. The unmixing loss ensures accurate spectral reconstruction. The background reconstruction loss is designed for background reconstruction and anomaly suppression. The smoothness loss enforces physically plausible endmember, and the orthogonality loss encourages distinctness among endmembers by promoting a near-orthogonal endmember matrix.

Given a hyperspectral image $\mathbf{Y} \in \mathbb{R}^{B\times N}$ (B bands and N pixels), the network generates each pixel $\mathbf{y}_p$ as $\hat{\mathbf{y}}_p = \mathbf{A}_p\mathbf{E}$, where $\mathbf{A}_p \in \mathbb{R}^{1\times M}$ is the abundance vector and $\mathbf{E} \in \mathbb{R}^{M\times B}$ is the endmember matrix. The reconstruction error for a single pixel combines spectral angle distance (SAD) and mean squared error (mse)

$$\mathrm{RE} = \arccos\left(\frac{\langle \mathbf{y}_p, \hat{\mathbf{y}}_p\rangle}{\|\mathbf{y}_p\| \cdot \|\hat{\mathbf{y}}_p\|}\right) + \|\mathbf{y}_p - \hat{\mathbf{y}}_p\|^2. \tag{11}$$

The unmixing loss $\mathscr{L}_{\mathrm{RE}}$ is the weighted average of the RE for all pixels. In addition, to obtain spectrally smooth and physically plausible endmembers, we apply a total variation (TV) regularization along the spectral dimension

$$\mathscr{L}_{\mathrm{MTV}} = \frac{1}{M}\sum_{m=1}^{M}\frac{1}{B-1}\sum_{b=1}^{B-1}\left|\mathbf{E}_{m,b+1} - \mathbf{E}_{m,b}\right|. \tag{12}$$

This penalizes large differences between adjacent bands, reducing noise-induced oscillations. In addition to the main reconstruction branch, our network includes a dedicated background reconstruction branch that learns a compact representation of the background. This branch receives the same input $\mathbf{Y}$ but is designed to reconstruct background pixels accurately while failing on anomalies [55]. The input and reconstructed hyperspectral vectors are denoted as $\mathbf{y}_p$ and $\tilde{\mathbf{y}}_p$. Consistent with the anomaly detection method based on background reconstruction learning [55], [60], [67], for the loss optimization of the network, our network is trained using the mse loss function as follows:

$$\mathscr{L}_{\mathrm{BG}} = \frac{1}{|\mathbf{Y}|}\sum_{\mathbf{y}_p\in\mathbf{Y}}\|\mathbf{y}_p - \tilde{\mathbf{y}}_p\|^2. \tag{13}$$

This loss encourages the background branch to capture the statistical regularities of the background, and the reconstruction residual from this branch ($\|\mathbf{y}_p - \tilde{\mathbf{y}}_p\|$) serves as an additional anomaly indicator. To ensure that the learned endmembers are mutually distinct and avoid redundancy, we impose an orthogonality constraint on the endmember matrix. Following the classic VCA mechanism, we encourage the columns of the endmember matrix to be nearly orthogonal. Specifically, we define the orthogonality loss, which minimizes the squared cosine similarity between different endmembers. Alternatively, when a learnable endmember matrix $\mathbf{E}$ is utilized, the loss can be expressed as

$$\mathscr{L}_{\mathrm{ortho}} = \left\|\mathbf{E}^{\top}\mathbf{E} - \mathbf{I}\right\|_F^2 \tag{14}$$

where $\mathbf{I}$ represents the identity matrix. This formulation directly penalizes nonorthogonality. The total loss is a weighted sum of all components

$$\mathscr{L}_{\mathrm{total}} = \mathscr{L}_{\mathrm{RE}} + \lambda_{\mathrm{BG}}\mathscr{L}_{\mathrm{BG}} + \lambda_{\mathrm{MTV}}\mathscr{L}_{\mathrm{MTV}} + \lambda_{\mathrm{ortho}}\mathscr{L}_{\mathrm{ortho}} \tag{15}$$

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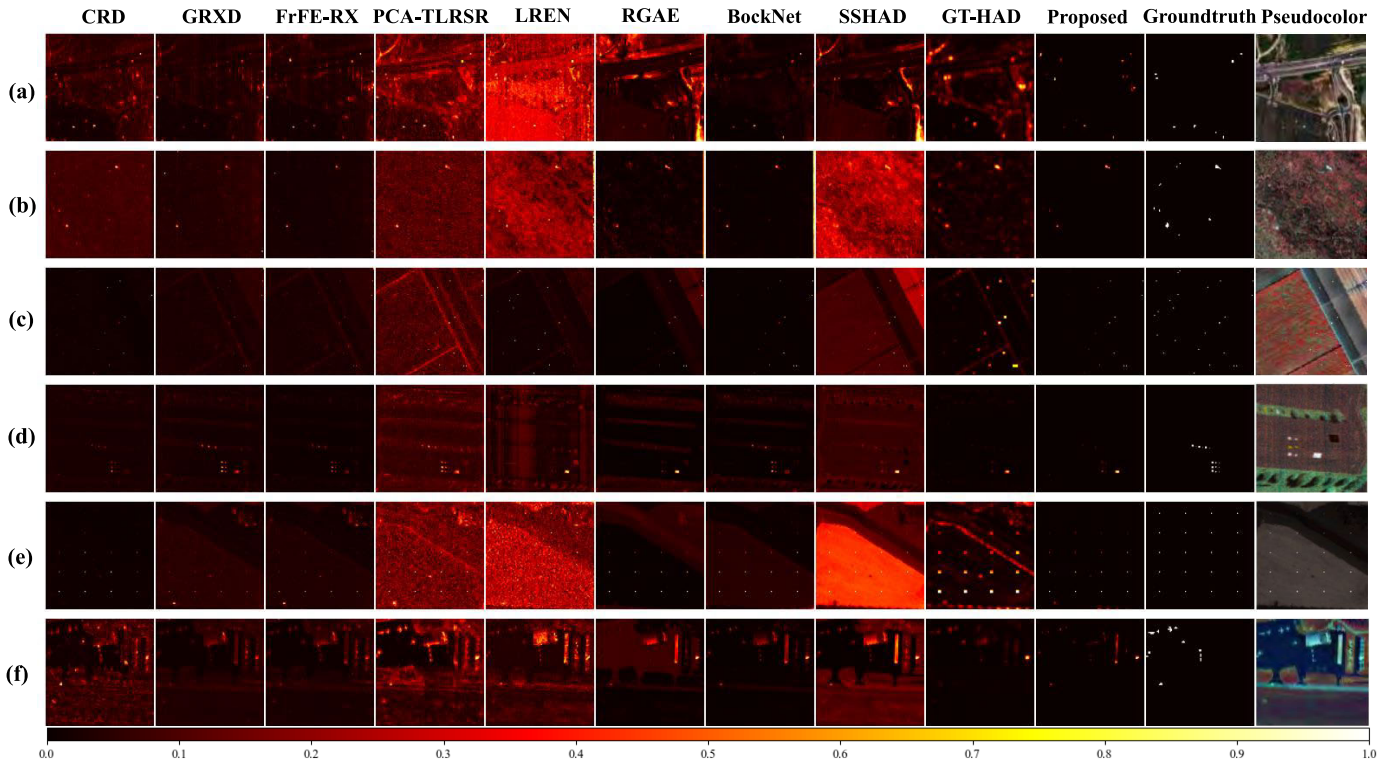


Fig. 7. Pseudocolor map, ground truth, and anomaly response maps of ten HAD algorithms on six HSI datasets. (a) HYDICE. (b) Hyperion. (c) Salinas. (d) SpecTIR. (e) Syntheticdata. (f) Gainesville.

where $\lambda_{\mathrm{BG}}, \lambda_{\mathrm{MTV}}$, and λ_{ortho} are hyperparameters controlling the contribution of each term.

IV. Experimental Results and Analyses

A. Experimental Platform

Extensive evaluations are performed on six HSI datasets to assess the effectiveness and robustness of the proposed JODINet for HAD and HU tasks. All implementations were executed on a Linux-based workstation equipped with an Intel[1] Core[2] i7-9750H CPU@2.60 GHz, 16 GB of RAM, and an NVIDIA 3090 GPU with 24 GB for accelerated computation within a Python environment.

B. Data Description

We benchmark our JODINet against current state-of-the-art detection methods using six widely adopted HSI datasets. The last two columns of Fig. 7 display the pseudocolor representations alongside their corresponding ground-truth annotations for each dataset. It is worth noting that the anomaly instances across these scenes exhibit considerable variation in both spatial extent and object category, thereby enabling a thorough evaluation of JODINet's detection capability across targets of disparate scales and types.

1) HYDICE Dataset: The first dataset comprises an urban hyperspectral image acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor [50]. This dataset covers a spectral range of 450–1350 nm with a 10-nm spectral resolution and contains 175 retained spectral bands. With a spatial resolution of 1 m/pixel and dimensions of 80×100 pixels, the scene primarily features cars and rooftops as anomalous targets against an urban background. As depicted in Fig. 7(a), a total of 21 anomalous pixels are present, constituting approximately 0.26% of the entire dataset.

2) Hyperion Dataset: The second dataset is a terrestrial hyperspectral image acquired by the Hyperion sensor aboard the EO-1 satellite over the Okavango Delta region [72]. Spanning a spectral range from 400 to 2500 nm, the dataset retains 145 spectral bands to preserve discriminative information for each pixel. The image measures 100×100 pixels with a spatial resolution of 30 m/pixel, and the background scene encompasses 14 distinct land-cover classes. As shown in Fig. 7(b), the dataset contains merely 32 anomalous pixels, accounting for 0.32% of the total pixel population.

3) Salinas Dataset: The third dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) hyperspectral sensor over the Salinas Valley in CA, USA [73]. It comprises 204 spectral bands spanning the wavelength range of 400–2500 nm. The image measures 120×120 pixels with a spatial resolution of 3.7 m/pixel. As illustrated in Fig. 7(c), the scene contains 25 anomalous pixels, which primarily consist of synthetic targets embedded within an agricultural background.

4) SpecTIR Dataset: The fourth dataset [53] originates from the SpecTIR hyperspectral airborne experiment conducted over Rochester, New York. A 100×100 pixel subscene was extracted for experimentation, encompassing 120 spectral bands with a spectral sampling interval of 5 nm and a spatial

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resolution of 1 m/pixel. Anomalous targets in this scene consist of artificially colored square textile panels, while the predominant background materials are soil and vegetation. The false-color composite and corresponding ground-truth map are displayed in Fig. 7(d).

5) Syntheticdata Dataset: Fig. 7(e) displays the syntheticdata dataset employed in this study [52]. The background scene was extracted from an authentic hyperspectral image acquired by the AVIRIS over San Diego Airport. The original imagery spans a spectral range of 0.37 to 2.51 μm with a 10-nm spectral resolution and a 3.5-m ground sampling distance, measuring 400×400 pixels. Prior to background extraction, channels affected by low signal-to-noise ratio and water vapor absorption (namely, bands 1–6, 33–35, 97, 107–113, 153–166, and 221–224) were discarded, leaving 189 spectral bands for further processing. A 100×100 pixel subregion was subsequently cropped to serve as the background canvas. Anomalous targets were then introduced into this background via a linear implantation strategy formulated

$$\boldsymbol{g} = \alpha \times \boldsymbol{t} + (1 - \alpha) \times \boldsymbol{b} \tag{16}$$

where $\boldsymbol{g}$ denotes the synthesized anomaly pixel, $\boldsymbol{t}$ is the pure target spectrum, $\boldsymbol{b}$ represents the original background pixel at the embedding location, and α controls the mixing proportion of the target. As illustrated in Fig. 7(e), a total of 20 isolated anomaly pixels are arranged in a 5×4 grid, with the mixing coefficients α set row-wise from the first to the fifth row as 0.2, 0.4, 0.6, 0.8, and 1.0.

6) Gainesville Dataset: This hyperspectral dataset was acquired by the AVIRIS over Gainesville, FL, USA [74]. Following the exclusion of bands severely affected by water absorption and low signal-to-clutter ratio (SCR), a total of 191 spectral channels were retained for experimentation. The dataset spans a spatial extent of 100×100 pixels with a ground sampling distance of 3.5 m/pixel. As shown in Fig. 7(f), the scene contains 52 anomalous pixels corresponding to ship targets, which account for 0.52% of the total pixel count.

C. Evaluation Indicators

1) HU Metrics: For the datasets with ground-truth of abundance and endmember, such as the HYDICE dataset, two commonly used evaluation metrics [21] are introduced, including root-mean-square error (RMSE) and SAD, which are calculated by

$$\mathrm{RMSE}_A = \frac{1}{P}\sum_{p=1}^{P}\sqrt{\frac{1}{R}\left\|\mathbf{A}_p - \hat{\mathbf{A}}_p\right\|^2} \tag{17}$$

$$\mathrm{SAD}_E = \frac{1}{R}\sum_{r=1}^{R}\arccos\left(\frac{\langle \mathbf{E}_r, \hat{\mathbf{E}}_r\rangle}{\|\mathbf{E}_r\| \cdot \left\|\hat{\mathbf{E}}_r\right\|}\right). \tag{18}$$

For most real HAD datasets where neither ground-truth abundance nor endmember is available, the pixel RE is adopted as the principal criterion

$$\mathrm{RMSE}_Y = \frac{1}{P}\sum_{p=1}^{P}\sqrt{\frac{1}{R}\left\|\mathbf{y}_p - \hat{\mathbf{y}}_p\right\|^2} \tag{19}$$

$$\mathrm{SAD}_Y = \frac{1}{P}\sum_{p=1}^{P}\arccos\left(\frac{\langle \mathbf{y}_p, \hat{\mathbf{y}}_p\rangle}{\|\mathbf{y}_p\| \cdot \left\|\hat{\mathbf{y}}_p\right\|}\right). \tag{20}$$

2) HAD Metrics: To evaluate the relative performance of different HAD methods, we employ a suite of metrics that capture both detection accuracy and computational efficiency. The running time of each algorithm across the six HSI datasets serves as the primary indicator of operational speed. For a comprehensive assessment of detection quality, we utilize both qualitative and quantitative measures. Qualitatively, we examine the anomaly response maps, the anomaly-background separation map, and the 3-D receiver operating characteristic (3-D ROC) curves [75]. The intensity of the colormap provides an intuitive visual cue regarding the strength of anomaly responses and the degree of background attenuation. The anomaly-background separation map [76] further elucidates the statistical relationship between the two classes by rendering their respective box plots. A wider interquartile gap between the background and anomaly boxes signifies superior discrimination between the target and clutter distributions. Complementing these, the 3-D ROC curve traces the trajectory of the detection probability P_d against the false alarm rate F_a as the segmentation threshold τ varies continuously within the interval $(0, 1)$. For any fixed threshold, P_d and F_a are defined as follows:

$$P_d = \frac{\mathrm{FA}}{\mathrm{TA}}, \quad F_a = \frac{\mathrm{FD}}{\mathrm{TP}} \tag{21}$$

where FA and TA stand for the number of true anomaly pixels identified by the respective detectors and the total number of true anomaly pixels in the HSI, respectively. The total pixel count within the HSI is designated as TP, while the number of false detections occurring below a specified threshold τ is recorded as FD. The full 3-D ROC curve serves as a qualitative descriptor of the tradeoff between anomaly detectability and background suppression inherent to a given algorithm. To extract quantitative benchmarks, the 3-D representation is marginalized along each axis, producing three planar 2-D ROC variants: (P_d, F_a), (P_d, τ), and (F_a, τ). The corresponding AUC values $\mathrm{AUC}_{(d,f)}$, $\mathrm{AUC}_{(d,\tau)}$, and $\mathrm{AUC}_{(f,\tau)}$ are then computed via numerical integration, each bounded within the interval $(0, 1)$. These scalar quantities are subsequently synthesized into higher level metrics, including the overall detection (OD) measure and the signal-to-noise probability ratio (SNPR), which offer concise summaries of algorithmic performance. The precise formulations and numerical bounds of these derived indicators are as follows:

$$\begin{cases} -1 \le \mathrm{AUC}_{od} = \mathrm{AUC}_{(d,f)} + \mathrm{AUC}_{(d,\tau)} - \mathrm{AUC}_{(f,\tau)} \le 2 \\ 0 \le \mathrm{AUC}_{snpr} = \dfrac{\mathrm{AUC}_{(d,\tau)}}{\mathrm{AUC}_{(f,\tau)}} \end{cases} \tag{22}$$

where the total accuracy of the detectors is evaluated based on $\mathrm{AUC}_{(d,f)}$ and AUC_{od}. The background suppression capability of a detector is quantified by $\mathrm{AUC}_{(f,\tau)}$ and AUC_{snpr}, whereas its anomaly detection sensitivity is reflected by $\mathrm{AUC}_{(d,\tau)}$. Notably, a lower value of $\mathrm{AUC}_{(f,\tau)}$ signifies superior background suppression.

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D. Parameter Tuning

For the experiments of HAD, nine excellent detectors are introduced to assess the performance of the proposed JODINet. These compared methods consist of four statistical distribution-based methods, i.e., CRD [50], GRXD [11], FrFE-RX [47], and PCA-TLRSR [54]; and five state-of-the-art deep learning-based methods, i.e., LREN [59], RGAE [56], BockNet [62], SSHAD [66], and GT-HAD [64]. Table I provides a complete summary of the parameter configurations and settings adopted for the nine comparative methods on the HYDICE dataset. For the remaining five datasets, the parameters of the comparison algorithms are then fine-tuned according to the preferred ranges reported in the original literature. To evaluate the unmixing fidelity of the proposed JODINet, we benchmark its performance against eight representative algorithms drawn from deep learning paradigms on the HYDICE dataset. These unmixing algorithms encompass five deep networks tailored for blind unmixing: SSAF [40], DAEN [35], CyCU-Net [39], NLAEU [36], MiSiC-Net [37], PGMSU [38], MSNet [41], and UADNet [21]. The detailed parameterization of all nine detectors is fully presented in Table I. In addition, we investigate four momentous parameters that are tuned in JODINet. P represents the size of the spatial patch for batch processing. The hyperparameters $\lambda_{BG}, \lambda_{MTV}$, and λ_{ortho} introduced in Section III-F govern the loss weighting scheme and consequently exert a direct influence on both HAD and HU accuracy. In the following experiments, when a specific parameter is examined, all remaining parameters are held constant at their predetermined optimal values. The resulting fluctuations of the $\text{AUC}_{(d,f)}$ metric across the six datasets with respect to each hyperparameter are depicted in Fig. 8.

1) Impact of P: The parameter P governs the spatial size of the input patch extracted via the sliding window, thereby defining the local receptive field for both feature extraction and background reconstruction. As reported in Fig. 8(a), the detection performance exhibits a dataset-dependent sensitivity to P. The Syntheticdata dataset achieves its optimal $\text{AUC}_{(d,f)}$ at $P = 3$, consistent with the small spatial footprint of its point-like implanted anomalies. The Gainesville, SpecTIR, and Salinas scenes attain peak performance at intermediate sizes of $P = 5$, $P = 5$, and $P = 9$, respectively, whereas the Hyperion and HYDICE datasets favor slightly larger patches of $P = 7$ and $P = 11$, suggesting that more complex backgrounds benefit from broader contextual aggregation. Notably, a consistent decline in $\text{AUC}_{(d,f)}$ is observed across all datasets when P exceeds 13, as excessively large patches dilute localized anomaly signatures and impair the fidelity of background estimation. Consequently, the optimal patch size P is selected individually for each dataset to align the receptive field with the intrinsic spatial scale of the anomalies and the surrounding background component.

2) Impact of λ_{BG}, λ_{MTV}, and λ_{Ortho}: The influence of three key hyperparameters λ_{BG}, λ_{MTV}, and λ_{ortho} on the detection performance across the six HSI datasets is summarized in Fig. 8(b)–(d). The parameter λ_{BG} governs the strength of the background reconstruction learning in the detection branch.

TABLE I
PARAMETER SETTINGS OF THE NINE COMPARISON HAD METHODS AND EIGHT HU METHODS ON THE HYDICE DATASET

Algorithms	Parameters
CRD	$(w_{in}, w_{out}) = (3, 5)$, $\lambda = 1e^{-6}$
GRXD	$(w_{in}, w_{out}) = (5, 7)$
FrFE-RX	$p = 300$, $j = 5$
PCA-TLRSR	$((d, \lambda, \lambda')) = (15, 0.06, 0.01)$
LREN	$\lambda = 1e^{-2}$
RGAE	$\lambda = a = 1e^{-3}$, $n_{hid} = 100$, $S = 150$
BockNet	$B_s = 7$
SSHAD	$n = 3$, $p = 3$, $m = 4$, $(\alpha, \beta) = (0.1, 1)$
GT-HAD	$Lr = 1e^{-3}$
SSAF	$J = 5$
NLAEU	$Lr = 1e^{-4}$
DAEN	$\mu = 0.3$, $\lambda = 0.2$
MiSiC-Net	$\lambda = 1e{-}1$, $Lr = 1e^{-2}$, $N_f = 256$, $K_s = (3 \times 3)$
CyCU-Net	$\beta = 0.5$, $\delta = 1e^{-6}$, $\gamma = 1e^{-2}$
PGMSU	$\lambda_1 = 0.1$, $\lambda_2 = 3$, $\lambda_3 = 7$, $J = 4$
MSNet	$\alpha = 0.1$, $\beta = 0.03$
UADNet	$K = 4$, $(\lambda_{MSE}, \lambda_{RE}, \lambda_{MTV}) = (0.5, 0.1, 0.1)$, $(\lambda_E, \lambda_C) = (1e^{-2}, 1e^{-4})$
JODINet	$P = 11$, $(\lambda_{BG}, \lambda_{MTV}, \lambda_{ortho}) = (1e^{-1}, 1e^{-3}, 1e^{-2})$

Sweeping the value of λ_{BG} from 10^{-4} to 10^{0} reveals that an intermediate setting of 10^{-1} consistently yields the highest $\text{AUC}_{(d,f)}$ scores for the majority of scenes. In contrast, lower values (from 10^{-4} to 10^{-3}) and the higher endpoint (10^{0}) lead to noticeable declines, indicating that a balanced background prior is essential for effective clutter suppression without oversmoothing discriminative features. Consequently, λ_{BG} is fixed at 10^{-1}. The second parameter, λ_{MTV}, controls the TV penalty imposed on the endmembers to promote smoothness. Across the range $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}\}$, the $\text{AUC}_{(d,f)}$ scores demonstrate considerable robustness. Only minor degradations are observed on the Salinas and SpecTIR datasets at the largest values. This suggests that the unmixing and detection pipelines are largely insensitive to this term, provided that it remains within a moderate range. We therefore select $\lambda_{MTV} = 10^{-3}$ as a conservative choice that ensures smooth endmember estimation without undue constraint. In contrast, the orthogonality constraint λ_{ortho} exhibits pronounced sensitivity. A modest value of 10^{-2} consistently attains peak performance across all six datasets, yet increasing λ_{ortho} to 10^{-1} or 10^{0} induces a sharp deterioration in $\text{AUC}_{(d,f)}$, particularly for the SpecTIR, Hyperion, and Salinas scenes. This is attributed to excessive decorrelation of the endmember signatures, which undermines the physical fidelity of the unmixing process and severely impairs background reconstruction. Accordingly, λ_{ortho} is set to 10^{-2}. In summary, the adopted configuration of $\lambda_{BG} = 10^{-1}$, $\lambda_{MTV} = 10^{-3}$, and $\lambda_{ortho} = 10^{-2}$ is motivated by a systematic evaluation of performance trends across diverse hyperspectral imagery and is employed consistently in all subsequent experiments. However, the current hyperparameter selection relies on manual tuning and dataset-specific heuristics, lacking an adaptive mechanism to automatically determine optimal values for new scenes. Future research will investigate automated strategies to enhance practicality.

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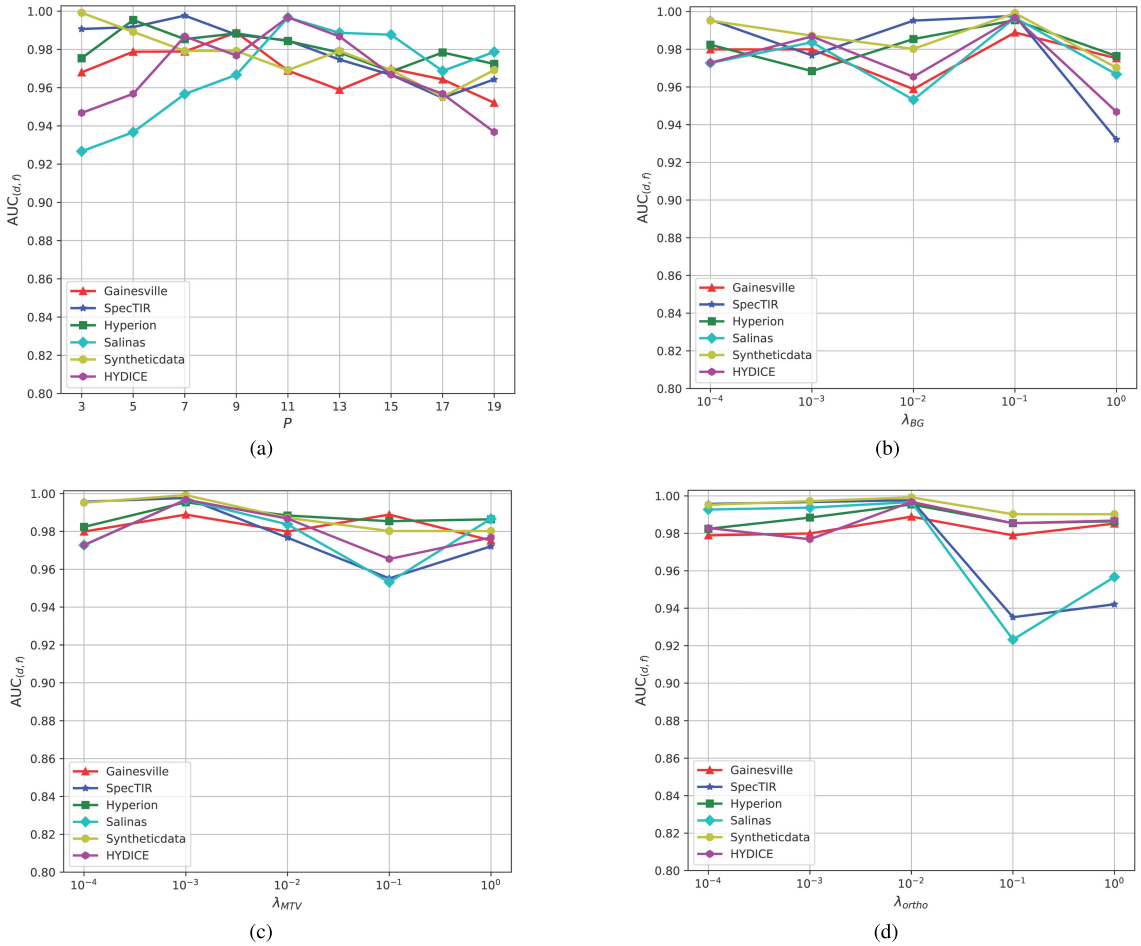


Fig. 8. Parameter tuning of the four key parameters of our JODINet on the six datasets. (a) Size of the spatial patch P. (b) Weight of background reconstruction loss λ_{BG}. (c) Weight of TV regularization loss λ_{MTV}. (d) Weight of endmembers orthogonality loss λ_{ortho}.

E. Detection Performance

This section provides a thorough comparative evaluation of the detection accuracy achieved by ten HAD algorithms across six HSI datasets.

1) Anomaly Response Maps: Fig. 7 displays the anomaly response maps of ten HAD algorithms on six HSI datasets, with color intensity reflecting detection confidence from dark to bright. Traditional detectors such as CRD, GRXD, PCA-TLRSR, and FrFE-RX generate widespread bright responses in background regions, indicating elevated false alarm rates across the HYDICE and Hyperion scenes. Other excellent methods, like RGAE and LREN, show modest improvement in anomaly localization but still retain noticeable residual clutter, particularly in the SpecTIR and Salinas datasets, where spectral variability challenges robust background estimation. Meanwhile, advanced deep learning approaches, including RGAE, BockNet, SSHAD, and GT-HAD, achieve strong background suppression. However, the response maps of these detectors become fragmented or subdued when confronting dense multiscale targets in the Synthetic and Gainesville scenes, often missing smaller anomalies entirely. In comparison, the proposed method consistently produces high-contrast bright regions that correspond precisely to ground-truth anomalies while maintaining a uniformly dark background response. This superior discriminability persists across all six scenes regardless of target scale, density, or scene complexity, confirming a more reliable balance between background suppression and anomaly preservation.

2) ROC Curves: As illustrated in Fig. 9, it provides a comprehensive visualization of detection performance through multiple ROC curve variants, including the 3-D (P_d, F_a, τ) representation and the corresponding 2-D (P_d, F_a), (P_d, τ), and (F_a, τ) ROC curve, evaluated across six distinct HSI datasets. For the conventional 2-D (P_d, F_a) ROC curve, proximity of the plotted ROC curve to the upper left corner is indicative of superior detection capability, as it signifies a higher probability of detection at a lower false alarm rate. A close inspection of the subfigures reveals that while several competing algorithms demonstrate moderate performance, their curves consistently lie further from the optimal corner, particularly in the low false alarm regime. In contrast, the curve corresponding to the proposed JODINet occupies the most favorable position across all five datasets, maintaining a pronounced elevation toward the upper left boundary. This advantage is further corroborated by the complementary 2-D ROC curve. In the 2-D (P_d, τ) ROC curve, the proposed method's curve consistently trends toward the upper right corner, signifying robust target detectability. Meanwhile, in the 2-D (F_a, τ) ROC curve, the

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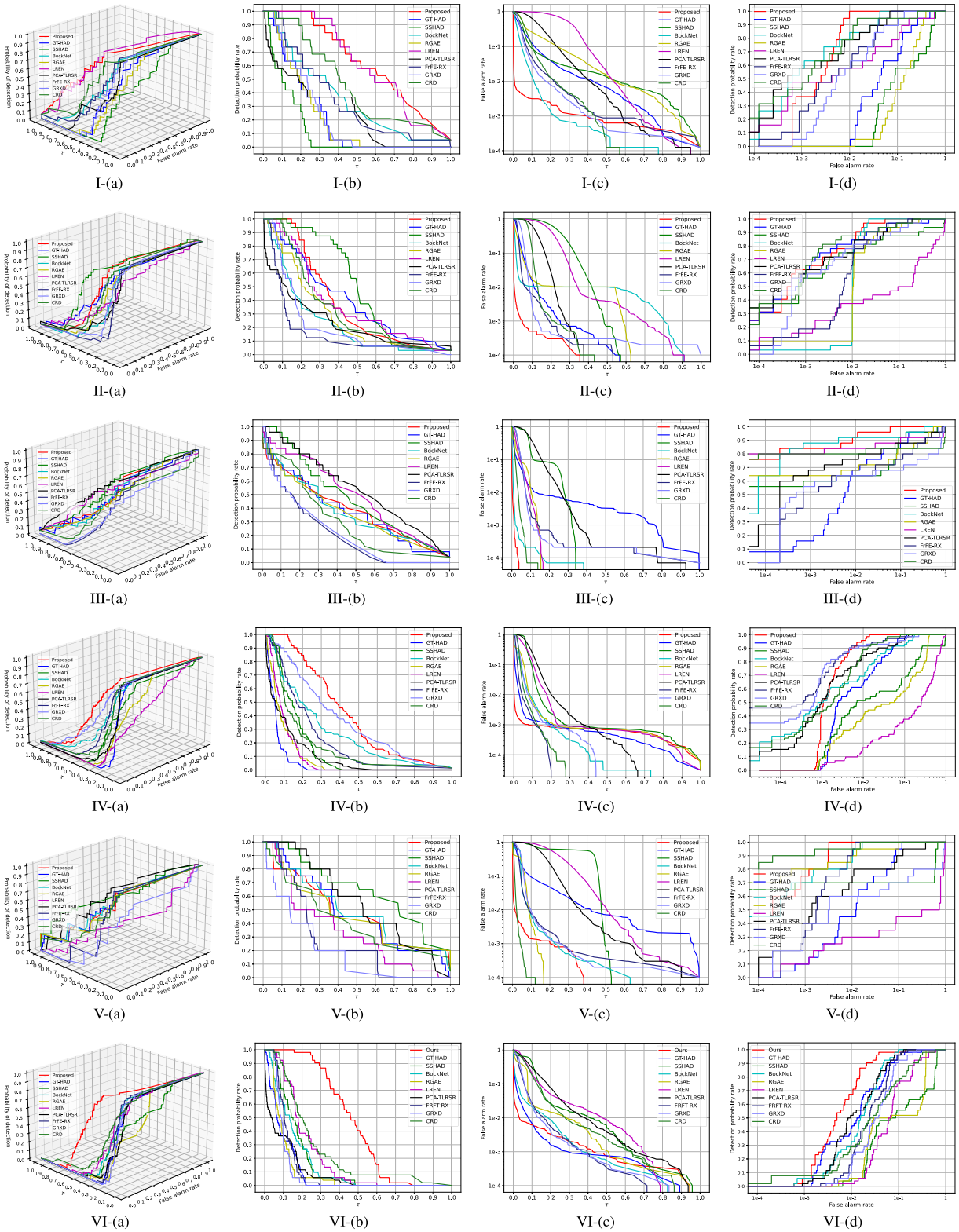


Fig. 9. I-(a): 3-D (P_d, F_a, τ) ROC curves on HYDICE dataset I-(b): 2-D (P_d, τ) ROC curves on HYDICE dataset I-(c): 2-D (F_a, τ) ROC curves on HYDICE dataset I-(d): 2-D (P_d, F_a) ROC curves on HYDICE dataset II-(a): 3-D (P_d, F_a, τ) ROC curves on Hyperion dataset II-(b): 2-D (P_d, τ) ROC curves on Hyperion dataset II-(c): 2-D (F_a, τ) ROC curves on Hyperion dataset II-(d): 2-D (P_d, F_a) ROC curves on Hyperion dataset III-(a):3-D (P_d, F_a, τ) ROC curves on Salinas dataset III-(b): 2-D (P_d, τ) ROC curves on Salinas dataset III-(c): 2-D (F_a, τ) ROC curves on Salinas dataset III-(d): 2-D (P_d, F_a) ROC curves on Salinas dataset IV-(a): 3-D (P_d, F_a, τ) ROC curves on SpecTIR dataset IV-(b): 2-D (P_d, τ) ROC curves on SpecTIR dataset IV-(c): 2-D (F_a, τ) ROC curves on SpecTIR dataset IV-(d): 2-D (P_d, F_a) ROC curves on SpecTIR dataset V-(a): 3-D (P_d, F_a, τ) ROC curves on Syntheticdata dataset V-(b): 2-D (P_d, τ) ROC curves on Syntheticdata dataset V-(c): 2-D (F_a, τ) ROC curves on Syntheticdata dataset V-(d): 2-D (P_d, F_a) ROC curves on Syntheticdata dataset VI-(a): 3-D (P_d, F_a, τ) ROC curves on Gainesville dataset VI-(b): 2-D (P_d, τ) ROC curves on Gainesville dataset VI-(c): 2-D (F_a, τ) ROC curves on Gainesville dataset VI-(d): 2-D (P_d, F_a) ROC curves on Gainesville dataset.

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TABLE II
AUC VALUES OF TEN HAD MODELS ON SIX HSI DATASETS

Data Sets	AUC value	CRD	GRXD	FrFE-RX	PCA-TLRSR	LREN	RGAE	BockNet	SSHAD	GT-HAD	JODINet
HYDICE	$AUC_{(d,f)}\uparrow$	0.9828	0.9763	0.9853	0.9896	0.9382	0.8371	0.9929	0.7799	0.9094	**0.9968**
	$AUC_{(d,\tau)}\uparrow$	0.4634	0.2184	0.3456	0.2501	0.6202	0.2102	0.3555	0.1220	0.2080	**0.6295**
	$AUC_{(f,\tau)}\downarrow$	0.0589	0.0380	0.0415	0.1372	0.2920	0.0948	0.0210	0.0572	0.0496	**0.0019**
	$AUC_{od}\uparrow$	1.3873	1.1566	1.2894	1.1025	1.2664	0.9525	1.3273	0.8447	1.0678	**1.6244**
	$AUC_{snpr}\uparrow$	7.8737	5.7404	8.3298	1.8228	2.1239	2.2178	16.8936	2.1328	4.1969	**331.3157**
Hyperion	$AUC_{(d,f)}\uparrow$	0.9884	0.9768	0.9849	0.9853	0.7392	0.9756	0.9891	0.9198	0.9808	**0.9954**
	$AUC_{(d,\tau)}\uparrow$	0.3333	0.2112	0.1610	0.2333	0.3783	0.2885	0.2305	**0.5046**	0.4016	0.3838
	$AUC_{(f,\tau)}\downarrow$	0.0811	0.0424	0.0232	0.1146	0.1902	0.0216	0.0130	0.2324	0.0254	**0.0007**
	$AUC_{od}\uparrow$	1.2406	1.1456	1.1227	1.1040	0.9273	1.2425	1.2066	1.1919	1.3570	**1.3785**
	$AUC_{snpr}\uparrow$	4.1101	4.9830	6.9427	2.0357	1.9886	13.3682	17.7263	2.1710	15.8040	**548.2857**
Salinas	$AUC_{(d,f)}\uparrow$	0.8464	0.8073	0.8977	0.9228	0.9164	0.9290	0.9812	0.9488	0.9511	**0.9967**
	$AUC_{(d,\tau)}\uparrow$	0.3053	0.2143	0.2090	**0.5415**	0.4903	0.3688	0.3723	0.4749	0.3803	0.3873
	$AUC_{(f,\tau)}\downarrow$	0.0108	0.0314	0.0305	0.1005	0.0239	0.0157	0.0017	0.0931	0.0147	**0.0002**
	$AUC_{od}\uparrow$	1.1409	0.9903	1.0762	1.3638	1.3828	1.2822	1.3519	1.3305	1.3167	**1.3838**
	$AUC_{snpr}\uparrow$	28.3988	6.8353	6.8511	5.3879	20.5113	23.5656	221.7635	5.0999	25.8610	**2251.8838**
SpecTIR	$AUC_{(d,f)}\uparrow$	0.9944	0.9914	0.9948	0.9895	0.6896	0.8776	0.9827	0.8655	0.9845	**0.9977**
	$AUC_{(d,\tau)}\uparrow$	0.1826	0.3683	0.2209	0.1017	0.0964	0.0891	0.2733	0.1573	0.0560	**0.4145**
	$AUC_{(f,\tau)}\downarrow$	0.0244	0.0249	0.0118	0.0795	0.0573	0.0176	0.0127	0.0690	0.0034	**0.0009**
	$AUC_{od}\uparrow$	1.1526	1.3348	1.2039	1.0117	0.7287	0.9491	1.2434	0.9538	1.0371	**1.4113**
	$AUC_{snpr}\uparrow$	7.4955	14.8174	18.6606	1.2792	1.6818	5.0502	21.6018	2.2798	16.2550	**460.5555**
Syntheticdata	$AUC_{(d,f)}\uparrow$	0.9991	0.7960	0.9968	0.9643	0.5412	0.9937	0.9975	0.8145	0.9732	**0.9992**
	$AUC_{(d,\tau)}\uparrow$	0.4195	0.1927	0.2733	0.5585	0.3413	0.4676	0.4955	**0.6458**	0.4890	0.4760
	$AUC_{(f,\tau)}\downarrow$	0.0076	0.0405	0.0299	0.1608	0.2336	0.0229	0.0312	0.3074	0.0471	**0.0015**
	$AUC_{od}\uparrow$	1.4111	0.9483	1.2403	1.3620	0.6488	1.4384	1.4618	1.1529	1.4151	**1.4737**
	$AUC_{snpr}\uparrow$	55.4917	4.7615	9.1514	3.4740	1.4606	20.3839	15.8997	2.1011	10.3787	**322.0663**
Gainesville	$AUC_{(d,f)}\uparrow$	0.9267	0.9513	0.9652	0.9765	0.8966	0.8200	0.9764	0.7663	0.9705	**0.9889**
	$AUC_{(d,\tau)}\uparrow$	0.2538	0.0963	0.1172	0.0890	0.2089	0.1007	0.1491	0.1655	0.0912	**0.4589**
	$AUC_{(f,\tau)}\downarrow$	0.0661	0.0351	0.0313	0.0862	0.0805	0.0377	0.0111	0.0809	0.0072	**0.0028**
	$AUC_{od}\uparrow$	1.1144	1.0125	1.0512	0.9793	1.0249	0.8831	1.1145	0.8509	1.0545	**1.4450**
	$AUC_{snpr}\uparrow$	3.8413	2.7443	3.7498	1.0324	2.5941	2.6747	13.4535	2.0463	12.6955	**163.8928**
Average	$AUC_{(d,f)}\uparrow$	0.9564	0.9164	0.9708	0.9697	0.7202	0.9055	0.9866	0.8491	0.9616	**0.9958**
	$AUC_{(d,\tau)}\uparrow$	0.3263	0.2169	0.2212	0.2957	0.3559	0.2541	0.3127	0.3450	0.2710	**0.5250**
	$AUC_{(f,\tau)}\downarrow$	0.0415	0.0354	0.0280	0.1131	0.1462	0.0351	0.0151	0.1400	0.0246	**0.0013**
	$AUC_{od}\uparrow$	1.2415	1.0980	1.1640	1.1539	0.9298	1.1247	1.2843	1.0541	1.2081	**1.4528**
	$AUC_{snpr}\uparrow$	17.8684	6.4804	8.1142	2.5053	5.0601	11.2102	51.2231	2.6385	14.1985	**679.6666**

curve remains anchored near the lower left region, reflecting markedly superior false alarm suppression. Collectively, the multiperspective ROC analysis confirms that the proposed model achieves a consistently higher detection rate while imposing lower responses on the background, underscoring its robust and reliable anomaly detection capabilities across diverse datasets.

3) AUC Values: To complement the qualitative visualizations, Table II reports the quantitative AUC metrics derived from the 3-D ROC analysis for all competing algorithms across six HSI datasets. Five complementary indicators are presented: $AUC_{(d,f)}$ and AUC_{od} (higher values denote superior OD), $AUC_{(d,\tau)}$ (reflecting target detection performance), $AUC_{(f,\tau)}$ (lower values indicate stronger false alarm suppression), and AUC_{snpr} (quantifying the signal-to-noise separation). The optimal result in each row is highlighted in bold, while the second-best entry is underlined. A cursory inspection of Table II reveals that the proposed JODINet consistently attains the highest or near-highest scores across nearly all datasets and metrics. JODINet secures the top rank on all six scenes for both comprehensive $AUC_{(d,f)}$ and AUC_{od} measures. It achieves average values of 0.9958 and 1.4528. These results yield substantial margins of 0.92% and 0.1685 over the second-best averages, recorded by BockNet and RGAE. Notably, for the background-focused $AUC_{(f,\tau)}$ metric, where lower scores denote stronger false alarm suppression, JODINet delivers an average of 0.0013, outperforming the nearest competitor (SSHAD at 0.0151) by an important factor and confirming its exceptional ability to suppress background clutter. This advantage is further magnified in the AUC_{snpr} indicator, where JODINet's average score of 679.67 eclipses the next best average (BockNet's 51.22) by more than an order of magnitude, underscoring the model's profound discriminative power between anomalies and background. Additionally, JODINet achieves optimal $AUC_{(d,\tau)}$ on five of the six datasets, with an average of 0.5250, exceeding all competitors. Collectively, these results across five distinct AUC metrics demonstrate that the proposed method sets a new benchmark, consistently surpassing both conventional detectors and modern DNN-based architectures in target detectability, background suppression, and anomaly separability.

4) Anomaly-Background Separation Maps: To further quantify the discriminative capability between anomalies and background, Fig. 10 employs anomaly-background separation

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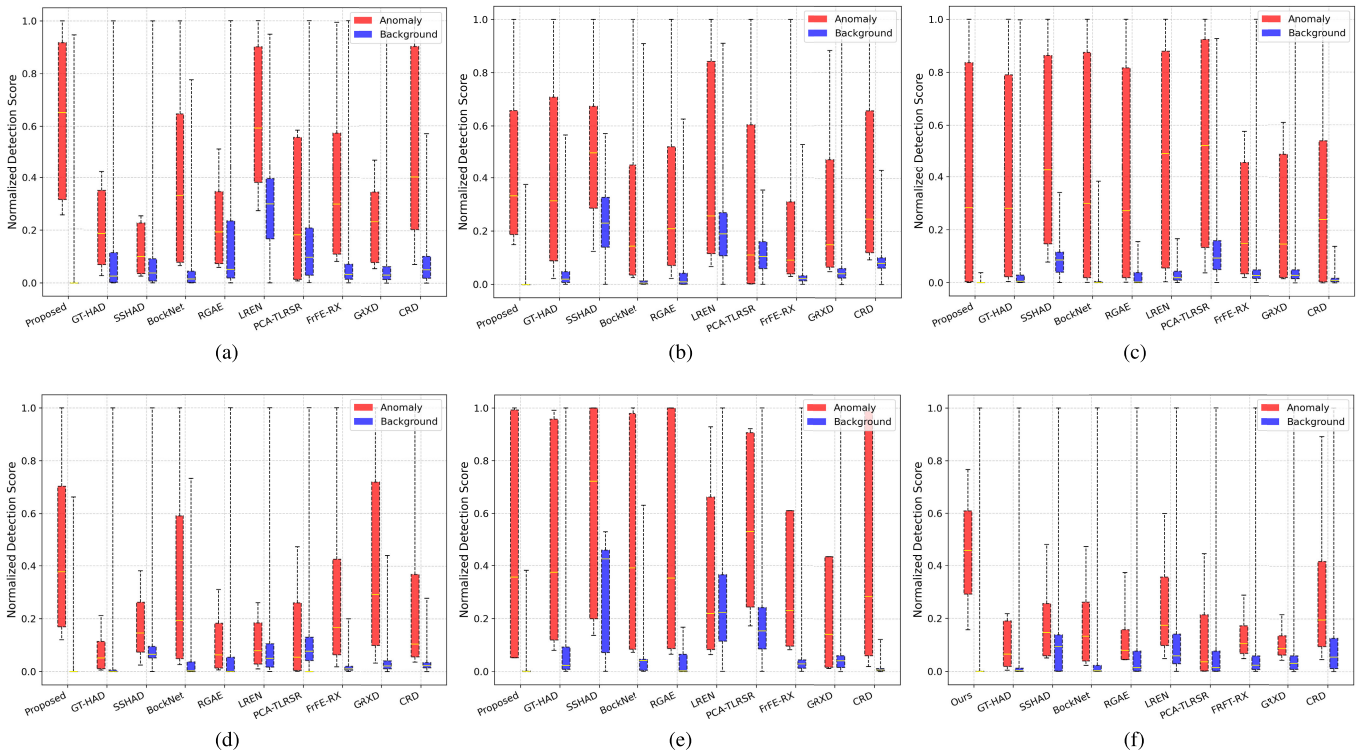


Fig. 10. Anomaly-background separation maps of the ten algorithms on six HSI datasets. (a) HYDICE. (b) Hyperion. (c) Salinas. (d) SpecTIR. (e) Syntheticdata. (f) Gainesville.

TABLE III
UNMIXING PERFORMANCE OF NINE HU ALGORITHMS ON SIX HSI DATASETS

Endmembers	Indicators	SSAF	NLAEU	DAEN	MiSiC-Net	CyCU-Net	PGMSU	MSNet	UADNet	JODINet
Hyperion	$RMSE_Y$	0.3112	0.2432	0.1524	0.4104	0.1027	0.3354	0.2431	0.1278	**0.0874**
	SAD_Y	0.0911	0.1562	0.2657	0.1289	0.0810	0.2150	0.2541	0.1124	**0.0524**
Salinas	$RMSE_Y$	0.3011	0.1898	0.1445	0.2356	0.2879	0.2431	**0.0991**	0.2132	0.1123
	SAD_Y	0.2211	0.3810	0.2470	0.2212	0.3109	0.2077	0.0970	0.1234	**0.0640**
SpecTIR	$RMSE_Y$	0.1456	0.3640	0.2897	0.2256	0.3214	0.1998	0.1277	0.1245	**0.1002**
	SAD_Y	0.1786	0.3321	**0.0767**	0.6543	0.5643	0.4544	0.0902	0.1776	0.0833
Syntheticdata	$RMSE_Y$	0.3344	0.4356	0.2898	0.3321	0.0734	0.1421	0.2865	0.1267	**0.0522**
	SAD_Y	0.3989	0.6643	0.1887	0.4377	0.1466	0.3381	0.4512	0.1298	**0.1064**
Gainesville	$RMSE_Y$	0.4231	0.2388	0.2890	0.3329	0.1990	0.2018	0.3354	0.1512	**0.1454**
	SAD_Y	0.5012	0.5539	0.6302	0.4875	0.4187	0.2689	0.3277	0.2211	**0.2139**
HYDICE	$RMSE_A$	0.2860	0.2540	0.2200	0.3020	0.1760	0.2440	0.2130	0.1572	**0.1490**
	SAD_E	0.2260	0.4430	0.1780	0.4330	0.4170	0.3500	0.1450	0.1187	**0.0983**

maps with box-whisker plots to illustrate the distributional separation across different HAD methods. In these diagrams, the red box encapsulates the statistical spread of anomaly response scores, whereas the blue box delineates the distribution of background pixel responses. A wider interquartile gap between the two boxes signifies superior separability, enabling clearer distinction of targets from the scene. Moreover, the vertical extent of the blue box directly reflects the degree of background suppression. A more compact, lower positioned blue box indicates minimal residual interference in nonanomalous regions. As depicted in the comparison, several competing algorithms exhibit a notable tradeoff between these two metrics. While certain methods (such as SSHAD and LREN) can improve the anomaly response (red box), they often do so at the expense of a tall or widely scattered blue box, leading to higher background response. Conversely, other approaches aggressively flatten the background distribution but compress the separation gap, thereby obscuring genuine anomalies. In contrast, the statistical profile of the proposed method reveals both a pronounced vertical gap between the red and blue boxes and a markedly reduced spread in background scores. This dual achievement confirms that the proposed model not only suppresses background clutter with high fidelity but also preserves a robust, high-contrast response for anomalous pixels across the evaluated HSI scenes.

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TABLE IV
UNMIXING PERFORMANCE OF NINE HU ALGORITHMS ON HYDICE DATASET

Endmembers	Indicators	SSAF	NLAEU	DAEN	MiSiC-Net	CyCU-Net	PGMSU	MSNet	UADNet	JODINet
Endmember1	RMSE	0.2960	0.2390	0.3220	0.3120	0.2150	0.3150	0.3130	0.2162	**0.2080**
	SAD	**0.0470**	0.1180	0.2820	0.1840	0.1170	0.0710	0.2240	0.1546	0.0730
Endmember2	RMSE	0.2900	0.2610	0.1490	0.2830	0.2590	0.2850	**0.1180**	0.1462	0.1380
	SAD	0.2270	0.3810	0.2470	0.2730	0.3450	0.3020	0.2370	0.2356	**0.1540**
Endmember3	RMSE	0.1890	0.2640	0.1360	0.3010	0.1120	0.1880	0.1450	0.1102	**0.1020**
	SAD	0.1930	0.6970	**0.0720**	0.7130	0.6960	0.7210	0.0750	0.1812	0.0996
Endmember4	RMSE	0.3670	0.2520	0.2750	0.3140	**0.1200**	0.1870	0.2760	0.1562	0.1480
	SAD	0.4380	0.5740	0.1110	0.5610	0.5100	0.3050	**0.0420**	0.1480	0.0664
Average	$RMSE_A$	0.2860	0.2540	0.2200	0.3020	0.1760	0.2440	0.2130	0.1572	**0.1490**
	SAD_E	0.2260	0.4430	0.1780	0.4330	0.4170	0.3500	0.1450	0.1187	**0.0983**

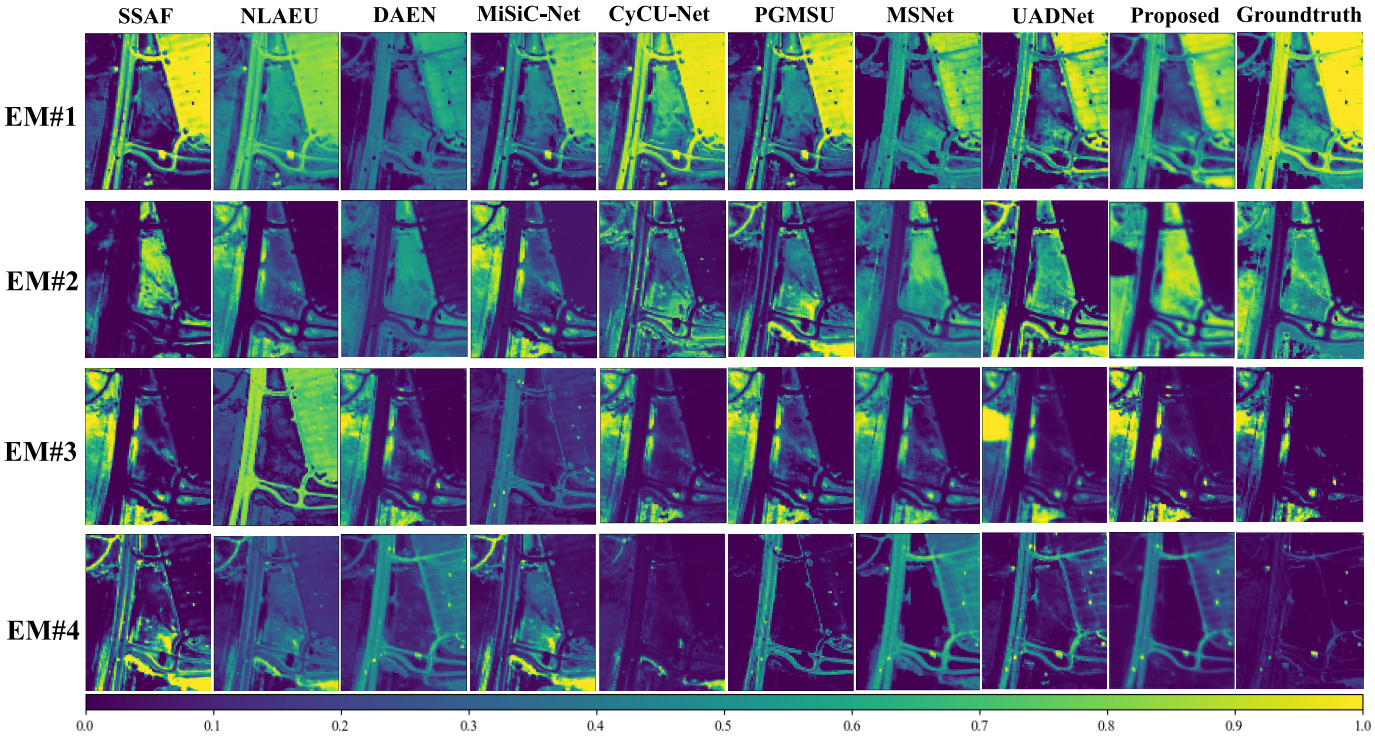


Fig. 11. Visual abundance maps of different methods on the HYDICE dataset.

F. Unmixing Performance

We conducted a comparative experiment on unmixing performance using the HYDICE dataset. For the proposed method, the hyperparameters P, λ_{BG}, λ_{MTV}, and λ_{ortho} are set according to Fig. 8. The hyperparameters of other comparison methods are set according to their original descriptions in Table I. In Table III, we report the unmixing performance of nine HU algorithms on six datasets, evaluated using two types of metrics mentioned in Section IV-C. For the HYDICE dataset, which provides ground-truth endmembers and abundances, we adopt $RMSE_A$ and SAD_E. For the other five datasets (Hyperion, Salinas, SpecTIR, Syntheticdata, and Gainesville datasets) where such ground truth is unavailable, the pixel reconstruction error ($RMSE_Y$ and SAD_Y) serves as the primary criterion, and the HYSIME algorithm [77] is first applied to estimate the number of endmembers. Among all algorithms, JODINet consistently achieves the best or competitive results on most datasets. Specifically, on the Hyperion dataset, JODINet yields the lowest $RMSE_Y$ (0.0874) and SAD_Y (0.0524). On the Salinas dataset, MSNet achieves the best $RMSE_Y$ (0.0991), while JODINet attains the best SAD_Y (0.0640). On the SpecTIR dataset, JODINet again gives the lowest $RMSE_Y$ (0.1002), whereas DAEN slightly outperforms in SAD_Y (0.0767 versus JODINet's 0.0833). For the Syntheticdata and Gainesville datasets, JODINet shows the best performance across both metrics (e.g., 0.0522/0.1064 and 0.1454/0.2139, respectively). On the HYDICE dataset, JODINet also achieves the smallest $RMSE_A$ (0.1490) and SAD_E (0.0983). Overall, JODINet exhibits superior robustness across diverse scenarios, while other algorithms, such as MSNet and DAEN, excel in specific cases. Specifically, for the HYDICE dataset with ground truth of endmembers and abundances, we also record the specific RMSE and SAD of

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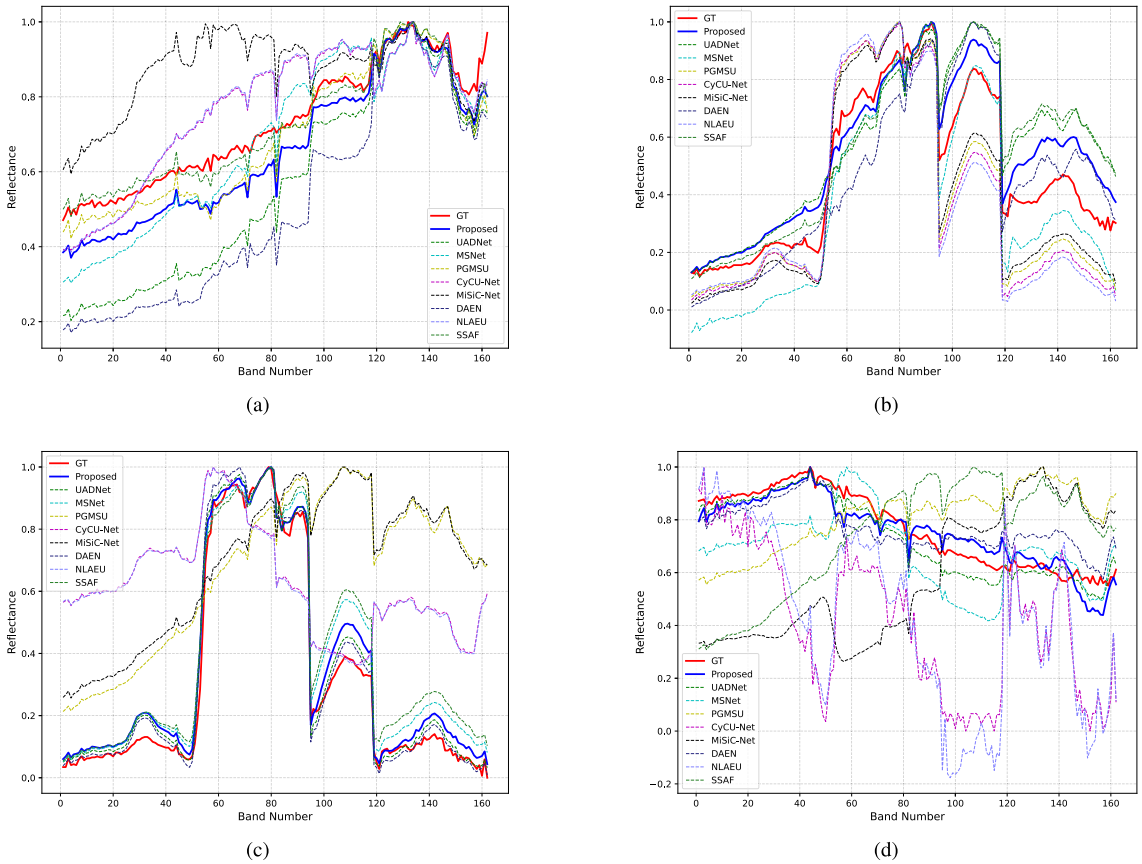


Fig. 12. Endmember estimation of different methods on the HYDICE dataset. (a) Endmember 1. (b) Endmember 2. (c) Endmember 3. (d) Endmember 4.

each endmember for more detailed and precise analysis in Table IV. The proposed JODINet achieves the lowest overall RMSE_A (0.1490) and SAD_E (0.0983) among all methods, outperforming its nearest competitor, UADNet, by about 5.2% and 17.2%, respectively. For endmembers 3 and 4, where spectral variability, the natural change in spectral signature due to material differences or environmental effects, is significant, JODINet consistently provides the most accurate estimates, as shown by its superior SAD values.

In addition, the estimated abundance maps in Fig. 11 clearly support the quantitative findings. Conventional deep learning-based unmixing approaches, such as MiSiC-Net and PGMSU, often yield abundance fractions (the proportions of each material at a pixel) that appear diffuse or affected by background interference. In contrast, JODINet produces abundance maps with sharp target boundaries and clear morphological details. The anomaly-associated endmember abundance (the spectral signature of anomalous materials) distinctly highlights the target regions. This stronger contrast shows that the unmixing stage effectively isolates anomalous materials from the background, providing a more reliable basis for detection tasks.

With regard to endmember extraction, the spectral signatures recovered by JODINet, as visualized in the corresponding spectral plots in Fig. 12, remain highly consistent with those derived by mainstream unmixing algorithms. This consistency suggests that the proposed method does not sacrifice general unmixing accuracy in favor of anomaly enhancement. Unlike competing approaches that rely on fixed or externally initialized spectral signatures, JODINet incorporates a learnable endmember module that dynamically refines the endmember estimates throughout the training process. Consequently, the recovered endmember spectra closely track the ground-truth reference curves, exhibiting minimal spectral distortion and accurate absorption feature alignment. This close correspondence is particularly evident in the estimation of Endmember 3, where other algorithms display pronounced deviations in both overall shape and local spectral gradients, whereas JODINet maintains high congruence with the true material signature.

G. Ablation Study

To systematically isolate and evaluate the contributions of the core architectural innovations, we conduct a series of ablation experiments by progressively integrating the BSCM, the ABIM, and the ELM into a unified baseline framework. The model variants are designated as **Model-A** (baseline without any specialized modules), **Model-B** (baseline + BSCM), **Model-C** (baseline + BSCM + ABIM), and **Model-D** (the complete proposed architecture with all three modules activated). Detection performance, quantified by $\text{AUC}_{(d,f)}$ across six distinct HSI datasets, is reported in Table V, while the concurrent evaluation of detection accuracy and unmixing fidelity is detailed in Table VI.

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TABLE V
QUANTITATIVE EXPERIMENTAL RESULTS OF ABLATION STUDY ABOUT HAD PERFORMANCE

Indicators	BSCM	ABIM	ELM	HYDICE	Hyperion	Salinas	SpecTIR	Syntheticdata	Gainesville	Average
Model-A	✗	✗	✗	0.9535	0.9442	0.9214	0.9489	0.9641	0.9465	0.9464
Model-B	✓	✗	✗	0.9665	0.9785	0.9701	0.9797	0.9822	0.9658	0.9738
Model-C	✓	✓	✗	0.9815	0.9889	0.9755	0.9834	0.9915	0.9821	0.9838
Model-D	✓	✓	✓	0.9968	0.9954	0.9967	0.9977	0.9992	0.9889	0.9958

TABLE VI
QUANTITATIVE EXPERIMENTAL RESULTS OF ABLATION STUDY ABOUT HAD AND HU PERFORMANCE ON SIX DATASETS

Model Type	Average $AUC_{(d,f)}$	Average RMSE	Average SAD
Model-A	0.9464	0.1666	0.1589
Model-B	0.9738	0.1534	0.1422
Model-C	0.9838	0.1178	0.1221
Model-D	**0.9958**	**0.1078**	**0.1031**

The progressive enhancement in average $AUC_{(d,f)}$ underscores the functional efficacy of each module. The integration of BSCM in **Model-B** elevates the baseline average from 0.9464 to 0.9738 (with an average $AUC_{(d,f)}$ gain of 2.74% over **Model-A**), a notable gain that directly reflects the module's design objective. By computing query–key attention between pooled central queries and their surrounding background keys, BSCM explicitly reinforces background homogeneity while suppressing outlier spectra that deviate from local consistency, thereby preventing the background decoder from inadvertently reconstructing anomalous targets. The subsequent addition of ABIM in **Model-C** yields a further improvement to an average of 0.9838 (with an average $AUC_{(d,f)}$ gain of 1% over **Model-B**). The most substantial performance leap is observed upon the inclusion of ELM in the complete **Model-D**, which attains an average $AUC_{(d,f)}$ of 0.9958 (with an average $AUC_{(d,f)}$ gain of 1.2% over **Model-C**) and secures optimal or near-optimal scores on all six evaluation scenes. This pronounced gain validates the critical role of the learnable endmember mechanism. As described in Section III-E, ELM refines the initial VCA-derived endmembers through a series of linear transformations, spectral normalization, and a spectral attention gate, enabling the network to adaptively correct residual estimation errors. The improved spectral fidelity of the endmembers directly translates into more accurate abundance estimation, which in turn provides a stronger foundation for anomaly discrimination.

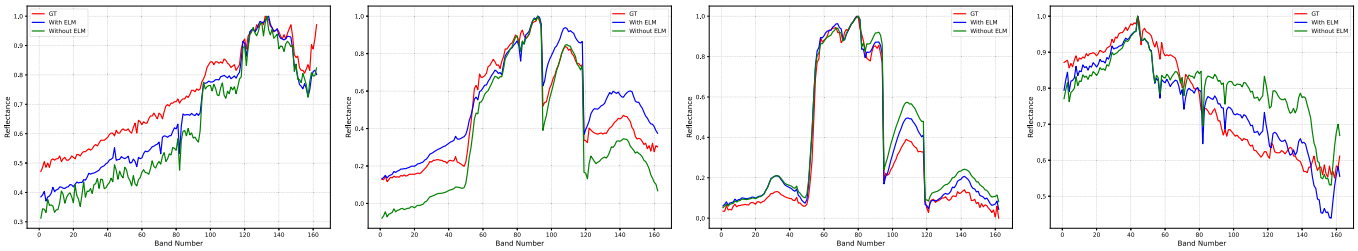
The synergy between average detection accuracy and unmixing precision of six datasets is further substantiated by the metrics in Table VI. There is a consistent downward trend in both RMSE and SAD as modules are sequentially activated. **Model-D** achieves the lowest reconstruction error (Average RMSE: 0.1078) and the smallest SAD (Average SAD: 0.1031). This parallel improvement confirms the value of physically constrained abundance estimation. Regularized by the attention-based interactions of BSCM and ABIM and refined by the adaptive endmember learning of ELM, this estimation not only guides feature learning but also yields representations more separable for anomaly detection.

Compared with **Model-B**, **Model-C**, which incorporates the ABIM module, achieves consistent improvements in both anomaly detection and unmixing performance. Specifically, the average $AUC_{(d,f)}$ increases from 0.9738 (**Model-B**) to 0.9838 (**Model-C**), while the average RMSE decreases from 0.1534 to 0.1178 and the average SAD drops from 0.1422 to 0.1221. These results demonstrate that the ABIM module effectively enhances the model's capability to separate anomalies and accurately unmix, leading to better overall HAD and HU performance. This increment is attributable to the bidirectional interaction mechanism established by ABIM, wherein abundance maps serve as queries to attend to background feature keys and values. The resulting gating operation enhances anomaly-sensitive abundance responses while simultaneously attenuating background regions contaminated by anomalous interference. Consequently, the joint refinement of abundance and background representations engenders a more discriminative feature space that directly benefits anomaly detection and unmixing. Notably, the transition from **Model-C** (without ELM) to the complete **Model-D** yields a pronounced reduction in unmixing error. RMSE decreases from 0.1178 to 0.1078 (a relative improvement of 8.5%), while SAD drops more substantially from 0.1221 to 0.1031, corresponding to a 15.6% reduction in spectral angular discrepancy. This marked improvement in SAD directly reflects the enhanced accuracy of endmember estimation conferred by the ELM, as the module adaptively refines the initial VCA-derived signatures to better align with true material spectra. Visual corroboration of ELM is provided in Fig. 13, which compares endmember signatures recovered with and without ELM. The curves obtained with ELM faithfully track the ground-truth reference across the spectral range, accurately learning spectral curve features. In contrast, variants lacking ELM exhibit substantial deviations, particularly in narrow spectral intervals where subtle material distinctions are paramount. This visual evidence reinforces the conclusion that the learnable endmember module effectively guides the unmixing process toward spectral representations, thereby underpinning the holistic performance gains documented throughout the ablation analysis.

H. Running Time

As reported in Table VII, we recorded the execution time (in seconds) of the ten competing algorithms across the six benchmark HSI datasets, with the average runtime across all scenes summarized in the final row. Note that the reported running times for all deep learning-based detection models (i.e.,

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(a) (b) (c) (d)

Fig. 13. Visual ablation study of the ELM on four endmembers. (a)–(d) Endmembers 1–4, respectively. The blue line denotes estimated endmembers with ELM. The green line denotes estimated endmembers without ELM. The red line denotes the ground truth of endmembers.

TABLE VII
RUNNING TIME (S) OF DIFFERENT HAD METHODS ON SIX DATASETS

Data sets	CRD	GRXD	FrFE-RX	PCA-TLRSR	LREN	RGAE	BockNet	SSHAD	GT-HAD	JODINet
HYDICE	25.8092	0.0571	20.2451	15.2451	38.5124	25.2414	97.3215	35.5412	28.2344	25.2451
Hyperion	27.3453	0.0623	23.5346	18.7867	43.5675	28.5785	104.6577	38.6785	30.5467	26.5463
Salinas	30.4235	0.0856	27.8678	23.2354	46.4564	32.4325	112.2344	41.2135	35.2342	27.3536
SpecTIR	28.5677	0.0343	25.2235	21.4576	38.3123	21.4542	101.7867	33.5464	27.1234	21.2342
Syntheticdata	31.4653	0.0887	32.2345	25.6786	44.7867	34.3466	121.5675	35.3465	31.7685	27.4564
Gainesville	26.3424	0.0633	21.5645	17.4564	35.8877	24.8567	98.4564	35.7898	27.4565	22.3453
Average	28.3256	0.0652	25.1117	20.3100	41.2538	27.8183	106.0040	36.6860	30.0606	25.0302

LREN, RGAE, BockNet, SSHAD, GT-HAD, and JODINet) include both the training phase and the inference phase. As expected, the classical statistical detector GRXD exhibits the fastest inference, averaging merely 0.0652 s per dataset due to its lightweight closed-form computation. Among the deep learning-based approaches, however, there exists substantial variability in computational demand. Notably, BockNet incurs the highest average runtime (106.0040 s), rendering it considerably less practical for time-sensitive applications. In contrast, the proposed JODINet achieves an average runtime of 25.0302 s. Specifically, JODINet is approximately 4.2× faster than BockNet and also demonstrates a modest speed advantage over SSHAD (36.6860 s) and GT-HAD (30.0606 s). While traditional methods such as PCA-TLRSR (20.3100 s) and FrFE-RX (25.1117 s) offer marginally lower or comparable latencies, they do so at the expense of the substantial detection accuracy improvements demonstrated in the preceding sections. Importantly, JODINet maintains this competitive runtime profile consistently across all six datasets, with per-scene execution times varying within a narrow band (21.23–27.46 s), indicating stable computational efficiency in scene complexity. These results confirm that the architectural design of JODINet strikes an effective balance between detection-unmixing accuracy and practical computational efficiency.

V. CONCLUSION

This article introduced JODINet, an end-to-end network framework that establishes an effective mutual feedback mechanism to yield robust anomaly detection and unsupervised unmixing within a unified architecture. Specifically, the proposed JODINet jointly optimizes HU and anomaly detection through parallel spectral and background decoders with a shared encoder and a carefully crafted multiobjective loss. In addition, an ABIM and a learnable ELM are designed for bidirectional cross-task refinement and precise endmember estimation, respectively. Extensive experiments on multiple real-world HSI datasets demonstrate that JODINet consistently achieves state-of-the-art detection accuracy and unmixing precision. In future work, we will focus on developing lightweight model variants and exploring efficient inference strategies to facilitate real-time HAD and unmixing without sacrificing discriminative capability. In addition, future research will investigate automated strategies such as Bayesian optimization or learnable hyperparameter modules to enhance practicality. Furthermore, the limited availability of joint task datasets for HAD and HU remains a bottleneck. We also plan to collect and publicly release new datasets for the HAD and HU joint task.

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